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Asia Primer | Asia Pacific

A Field Guide to Korea's Capital Markets

We provide a one-stop shop for investors who are looking to understand more about Korea's bond, equity and FX markets, and the macroeconomic policy backdrop affecting asset prices.

Changes in global indices draw attention to Korea: The Korean government has been keen to get Korean Treasury Bonds (KTBs) included in the FTSE World Government Bond Index (WGBI) and Korean equities upgraded from MSCI EM to MSCI DM. Key announcements related to these index changes may come as soon as late 2024. We help investors prepare for this in advance by understanding Korea's macro backdrop and the fundamentals of the bond, equity and FX markets.

Hawkish direction for monetary & fiscal policy: Korea's monetary policy is set by the Bank of Korea while fiscal policy is determined by the Ministry of Economy and Finance. The BoK has a dual mandate of price and financial stability, and currently has a hawkish bias given uncertainty around inflation and financial imbalance. Korea's usual fiscal surplus has turned into a deficit since 2020 amid fiscal activism, but government debt-to-GDP remains at a healthy level. The government is seeking to implement the Fiscal Rule to sustain medium-term fiscal soundness.

Korea's government bond market is one of the largest in AXJ, with a size of ~US\$825 billion. Including other types of bonds, Korea's bond market stands at ~US\$2 trillion. Korea has both a nominal and inflation KTB market, but the latter is less liquid. Banks are the largest buyers of KTBs. Foreigners' share has also been increasing, and the Euroclearability of KTBs (scheduled for 1H24) could help that to rise further, and is one of the key watch points for WGBI inclusion.

KRW is one of the most actively traded currencies, with an average daily turnover of US\$142 billion. Offshore non-deliverable forwards (NDFs) account for a large part of that, due to offshore investors' limited access to the onshore KRW market, though this is expected to improve with the government's proposed reforms from 2H24. Trade flows are a key source of USD supply while financial flows, particularly in FDI and equities, have been key sources of KRW outflows.

Korea's equity market is highly cyclical: Korea's equity market cap stands at US\$1.6 trillion, with IT, Industrials, and Energy & Materials accounting for more than 60% of KOSPI's market cap. Korea has one of the highest proportions of cyclical stocks in the world and hence has a tendency to lead EM bull market runs. While foreigners and pension funds are major players, retail investor participation has picked up since 2020 and this has changed market dynamics.

What differentiates Korea from other equity markets includes: 1) The government's push to mitigate the "Korea Discount"; 2) Corporate business restructuring and holdco structures; 3) Disparate stock liquidity dynamics linked to short-selling and rule changes; and 4) Investment proclivities of retail investors.

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Executive Summary

Macroeconomic policy backdrop

Monetary policy: We delve into Korea's central policy framework against the backdrop of domestic capital market operations. In the first section, we introduce Korea's monetary policy structure and the policy target set by the Bank of Korea (BoK). We start out with a brief history of its establishment and outline the Bank's dual mandates on price and financial stability.

We discuss its inflation targeting system (currently a headline CPI inflation of 2%Y), and drivers of the domestic inflationary environment. We also touch on the transmission effect of monetary policy in Korea, and provide an introduction to the top policy decision-making body, the Monetary Policy Committee (MPC). Finally, we share our latest views on the macroeconomic backdrop and the path of the BoK's policy rate.

Fiscal policy: In the second section, we delve into Korea's fiscal policy framework. We introduce the role of Ministry of Economy and Finance (MoEF), Korea's top government agency in charge of economic policy direction, government finance, fiscal policy and regulations.

We discuss its purpose of ensuring macroeconomic and financial stability, conducting policy coordination and allocation of national resources. In particular, we focus on Korea's fiscal soundness as we review the current fiscal position and medium-term outlook on government debt. Finally, we end with discussions on supplementary budgets and the latest efforts by the government to sustain medium-term fiscal soundness through implementation of Fiscal Rule in Korea.

With the macroeconomic policy backdrop in mind, we provide more details about the fundamental setup of Korea's bond, FX and equity markets in the remaining sections of the report.

Bond market

Overview: Korea's bond market has a total size of KRW 2,709 trillion (US\$2 trillion), or 125% of GDP. The government bond market, at KRW 1,096 trillion, is the largest, accounting for 40% of the overall domestic bond market and 64% of total bond trading volume. Korea's government bond market is the largest within AXJ as a % of GDP, and third-largest in absolute USD terms. There is both a nominal and inflation KTB market, but the inflation bond market is small and not as actively traded.

Market participants: Banks are the largest holders of KTBs, owning 46% of total KTBs outstanding. Insurers are also sizeable holders, followed by pension funds and securities companies. Combined, these four investor types hold 91% of total KTBs outstanding. Foreign ownership of KTBs has also doubled since end-2009 to reach 21% of the market, with demand emanating mainly from medium- to long-term investors.

Foreigners' access: To trade domestic bonds, foreigners currently need to open an

account with an onshore custodian. The government has signed an agreement with Euroclear to reintroduce an omnibus account for KTBs, which would allow foreign investors to trade KTBs without opening a separate local account. This is currently expected to be rolled out in 1H24 and is important for the WGBI inclusion process.

Alternatives to KTBs: Bonds issued by the BoK (MSBs), state-owned companies (special bonds) and policy banks (policy bank bonds) are sometimes seen as alternatives to KTBs. MSBs offer similar yields to KTBs and have relatively high foreign participation, but their trading volume is only 12-15% of government bonds'. Special bonds offer the highest yield spread over KTBs but are not traded as actively as MSBs and bank bonds.

FX market

Overview: KRW is one of the most actively traded currencies across G10 and EM, with an average daily trading volume of US\$142 billion. FX forwards account for half of the total trading volume, as offshore investors' limited access to the onshore KRW market has made offshore non-deliverable forwards (NDF) a key trading tool for foreigners. The onshore spot FX market is usually open from 9:00am to 3:30pm KST on business days.

Flow of funds: Being one of the largest exporters in the world, Korea has run a current account surplus almost all the time since 1998, making trade flows a source of USD supply. Shipbuilders are a key USD supplier, while energy importers are a key USD buyer. On the financial flows side, outbound FDI, domestic investors' investment in foreign equities, and reserve assets have been key sources of KRW outflows. The National Pension Service (NPS) is a key USD buyer. We estimate KRW is currently slightly undervalued.

Improving foreigners' access: As part of its efforts to get Korea included in global bond and equity indices, the Korean government has proposed FX reforms including opening up the onshore interbank FX market to offshore financial institutions and extending the onshore market's trading hours from 2H24. The implementation of these will be key to monitor. The government has also eased some reporting/documentation requirements for FX transactions.

Equity market

Overview: Korea's equity market has total market capitalisation of KRW2,086 trillion (US\$1.6 trillion), which is ~97% of GDP. KOSPI (Korea Composite Stock Price Index) is the benchmark equity index, with major large-cap stocks listed in the KOSPI market. Given Korea's economic landscape, it is still highly reliant on exports from manufacturers, IT, Industrials, and Energy & Materials, which account for more than 60% of KOSPI's market cap.

Market participants: Foreigners own ~30% of the KOSPI market, with higher exposure to IT and Financials, while pension funds like NPS have played major roles among domestic institutions. Meanwhile, retail investors have a higher presence in KOSDAQ than in KOSPI, and overall retail participation has increased in both markets since the pandemic outbreak in 2020 thanks to copious liquidity and a low interest rate environment.

"Korea Discount": Compared to global peers, Korea has been trading at a significant discount for a long time, which is often referred to as the "Korea Discount". We believe there are three main reasons behind the valuation discount:

1) governance issues and lack of minority shareholder protection; 2) lower shareholder returns; and 3) low transparency and accessibility of information.

In order to address these issues and to achieve capital market advancement, the Korean government has announced several regulatory reforms including a dividend process reform, mandating appraisal rights to protect minority shareholders, and tighter disclosure requirements. We expect more measures to be announced in the coming months, but it remains to be seen whether these policy efforts can mitigate the "Korea Discount" and bring a secular re-rating story for the market.

Nuances of the Korean market: Aside from the policy initiatives to reduce the "Korea Discount", there are several other frameworks that have shaped Korea's equity market: 1) short-selling policy; 2) corporate business restructuring (i.e., holdco transitions, split-offs); 3) changing stock liquidity dynamics; and 4) investment proclivities of retail investors.

A partial short-selling ban remains in Korea's equity market, and there is no timeline at this juncture to a full implementation of short-selling.

Meanwhile, we have seen many cases of business restructuring including holdco transitions, split-offs and spin-offs that have had diverging impact on minority shareholder returns.

We are also seeing changing stock trading dynamics as we see increasing migration of stocks from KOSDAQ to KOSPI. To add, requirements on listed preferred shares have been tightened for the past few years to protect shareholders from low liquidity.

Retail investors have a greater tendency to seek capital gains in volatile markets and chase themes compared to institutional investors, which we believe is attributable to common behavioural biases.

Monetary Policy Framework

Kathleen Oh

The Bank of Korea

Founded in June 1950, the Bank of Korea (BoK) is the central bank of Korea, the top, independent monetary policy-making body. It was established on the legal foundation of the Bank of Korea Act, stating the purpose of the Bank in Article I, Clause I was "to contribute to the sound development of the national economy by pursuing price stability through the formulation and implementation of efficient monetary and credit policies". With the key objective to achieve sustainable economic growth, the Bank manages the monetary base and circulated liquidity in the domestic banking system to stabilize the overall economy's price level.

In addition to formulating monetary policy, the Bank performs traditional functions of a central bank: currency issuance, acting as the bank of last resort to local banks and the government, operating and overseeing the payment and settlement system, maintaining the foreign exchange reserves, and conducting supervisory functions for financial institutions.

Objectives of monetary policy

The BoK currently holds a dual mandate of price stability and financial stability. The Bank first implemented an inflation target to achieve price stability in 1997, which is specified as the most important objective of its policy decisions. In 2011, the Bank added financial stability as one of the two mandates of its policy operation. This was in accordance with the goal to strengthen the financial stability function of central banks globally post-Global Financial Crisis. Article I, Clause II of the Bank of Korea Act was added with the phrase that the BoK "shall pay attention to financial stability in carrying out its monetary and credit policies".

Key factors of monetary policy decisions are primarily the overall economic backdrop and growth outlook, followed by consumer price stability and private sector financial conditions. The trend in domestic growth activity momentum comes as a highly important variable as the BoK ultimately aims to promote overall development of the economy. Policymakers carefully balance between a sound and stable growth backdrop, and seeking the dual mandate of price and financial stability in the policy decision process.

Exhibit 1: Summary table of Bank of Korea

Summary	
Name	Bank of Korea
Headquarter	Seoul, South Korea
Established	12 June, 1950
Governor	Chang Yong Rhee
Decision-making body	Monetary Policy Committee (MPC)
Monetary Policy target	Headline CPI inflation at 2% on medium term
Policy committee structure	7 voting members (incl. governor)
Legal foundation	Bank of Korea Act

Source: Bank of Korea, Morgan Stanley Research

The BoK overall operates similarly to other major developed market central banks, as it targets CPI inflation and promotes financial stability. One distinctly different aspect from the US Federal Reserve is that the BoK does not have labour market mandate, such as a target unemployment rate. And unlike other central banks with reserve-currency status, it may be less free from having to defend the currency.

Inflation targeting

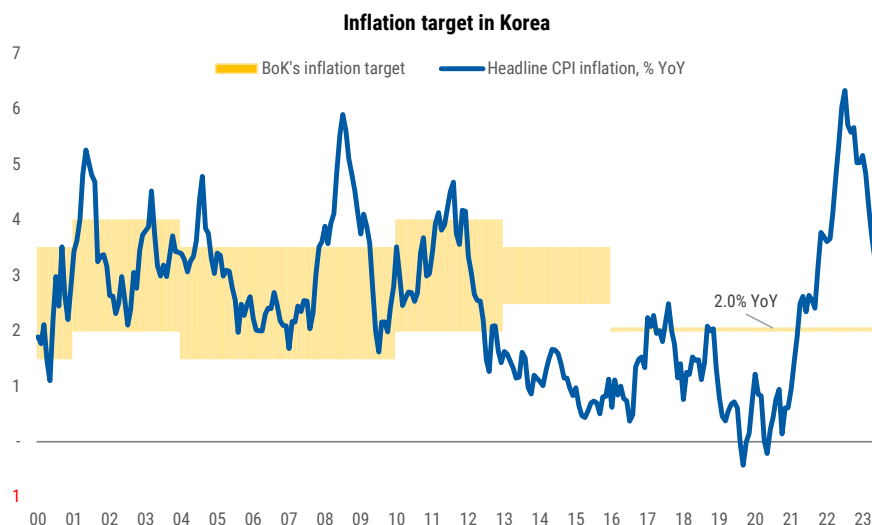
The BoK's policy formulation is based on an inflation targeting framework. Article 6, Clause I of the Bank's Act specified the inflation target for 2019 onwards has been set at 2.0%, the headline year-over-year change in the Consumer Price Index (CPI).

Since 2000, the Bank has revised inflation targets in accordance with the broader macroeconomic conditions. The Bank has pointed out that range-bound targets may increase uncertainty around optimal inflation levels and may destabilize inflation expectations. Before adopting a single-level inflation target at 2.0% from 2016, the Bank implemented a flexible range-bound target between 2.5-3.5% during 2013-16 with the mid-target level at 3% and range of +/- 50bps, and 2-4% in 2010-12 with the mid-target level at 3% and a range of +/- 100bp. Prior to the global financial crisis, it set the mid-target at 2.5% with a range of +/- 100bp of 1.5-3.5% between 2004 and 2009.

In selecting a target of 2.0%, the Bank noted that it saw Korea's optimal inflation around a CPI inflation of 2%, and assessed that Korea's underlying inflation had come down to 2% since 2012 due to demographics change, a drop in potential growth level, and weaker supply and demand-side inflationary pressure due to globalization.

Given the accountability for inflation targeting, the BoK holds a tight monitoring system. It conducts internal inflation target evaluations every two years in consultation with the government. The governor appears before the National Assembly on request to present the latest assessment on inflation, in addition to a biannual publication of the Monetary Policy Report, including an assessment of the latest price developments, inflation forecasts and risk factors to meeting the target.

Exhibit 2: Historical changes of BoK's inflation target and headline CPI inflation trend



Source: Bank of Korea, Morgan Stanley Research

The governor may hold a press conference to communicate an inflation assessment to the public as needed. In the case where inflation deviates from the 2.0% target by 50bps for six consecutive months, the BoK governor is requested to hold a press conference to explain and communicate factors driving the off-target inflation trend. If the deviation continues, additional sessions will be held on a quarterly basis.

With emerging discussions of adjusting current inflation targets globally, the Bank maintains a stance that it would "adjust the target in consultation with the government... if adjustment of the inflation target becomes inevitable, due for instance to unexpected domestic or global economic shocks or to changes in economic conditions..." ([BoK press release on inflation target review for 2019](#), December 26, 2018).

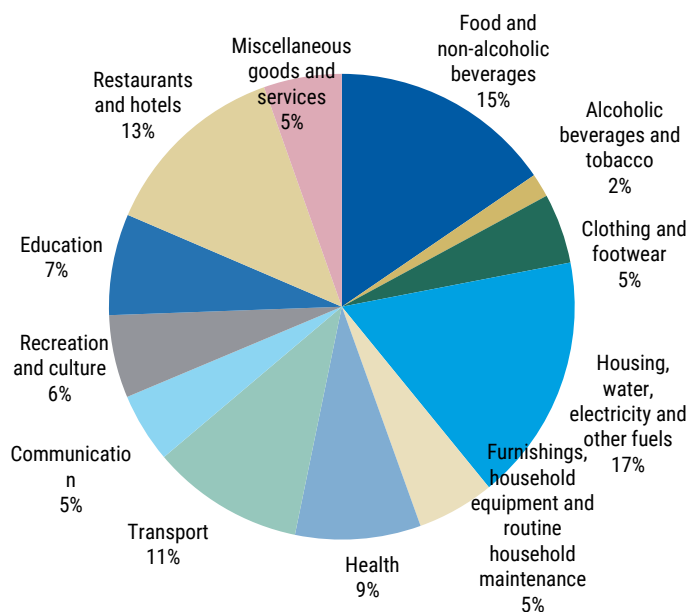
Key drivers and components of inflation

Drivers of inflation in Korea are well-balanced. The weight of each component in the CPI basket reflects their relative importance in household consumption. Weighting of commodities and services prices in Korea's CPI basket are about half each with commodities accounting for 46% of the basket and services accounting for 54%. Compared to other economies with a heavy dependence on one or two categories, i.e., UK with a 30% weighting on housing and India with 40% on food, no category exceeds a weight over 20% among the top 12 categories, which makes Korea's weighted average price move less vulnerable.

However, supply-side factors may drive large, visible changes at times as Korea relies on imported commodities. In particular, Korea relies entirely on imported energy. Global energy prices, especially crude oil and natural gas prices, can influence energy-related prices across transport (11% weighting in the total basket) and utilities (3%). Volatile moves in the Korean won (KRW) also affect imported price levels, often contributing to supply-side factors. Meanwhile, wage growth in general puts less force on inflation compared to other economies as it remains largely conservative due to rigidity of the labor market and heavy labor protection laws in Korea. Fiscal policy-driven inflation also usually remains limited as the government contains fiscal debt growth.

The key components of Korea's CPI basket are divided into 12 categories. Since 2020, the basket has been composed of total 458 goods and services products to gauge the overall changes in the economy's price level. The top components with the largest weighting are housing and utilities (17.1%), food and non-alcoholic beverages (15.4%) and restaurants and hotels (13.1%).

Korea rebalances its CPI basket every 5 years. While the big 12 categories are kept unchanged, there could be changes of around 5% to the total number of components depending on the products' relevance to the domestic consumption trends. For example, more frequently purchased items such as electric vehicles were added in the last rebalancing of the basket back in 2021, while less popular items such as printers and cameras were removed.

Exhibit 3: 12 components and weights of Korea's CPI basket

Source: Bank of Korea, Morgan Stanley Research

Policy rate and transmission effect

The BoK's primary tool to adjusting the stance of monetary policy is through its policy rate (also known as the base rate). The policy rate is the interest rate that the local financial institutions (FIs) pay to borrow reserve balances in Korea. It is a fixed bid for the 7-day repurchase agreements (RPs) and BoK's liquidity adjustment deposits and loans, which are 100bp below and above the policy rate, respectively. In order to determine the cost of short-term credit used by local FIs, the MPC sets the policy rate at each meeting. It has the most immediate transmission impact on the call rate, including the overnight interbank lending rate, and commercial banks' deposit rates and borrowing costs.

Historically, the BoK's policy rate tended to move similarly to other major central banks, as the global macro environment had a spillover effect onto the Korean economy – for example, when global oil price spiked last year, the strong USD led by a hawkish Fed resulted in hawkish BoK, along with other central banks in the region, namely RBA. And as importantly, due to the direct impact on USDKRW movement from the Fed, the BoK closely watches the policy direction in the US.

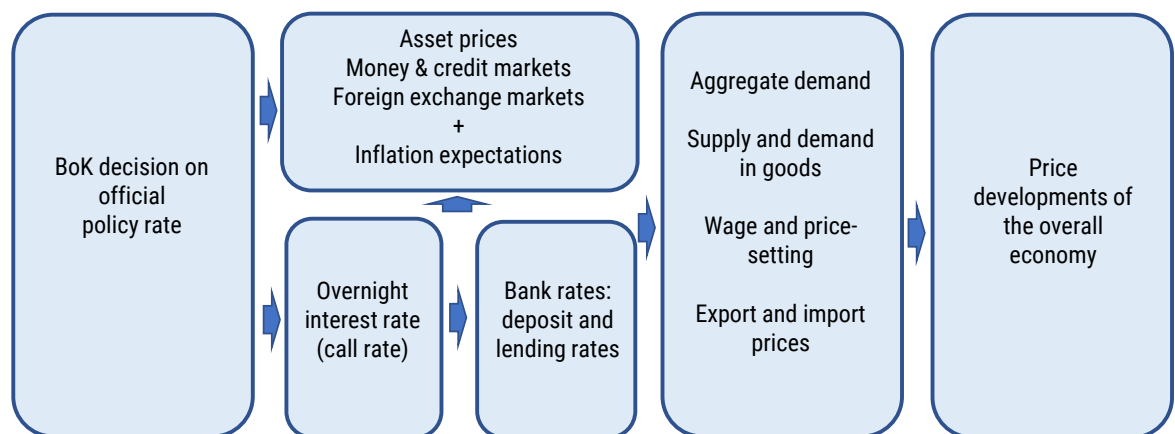
The BoK used to be more sensitive about keeping the rate differential with the US as tight as possible, or no wider than 150bps under the previous governorship (prior to 2022); but more recently, concerns around capital outflows due to a flipped rate level (a lower policy rate in Korea vs. the US) have receded, leading to a weaker linkage to the Fed or global banks' moves, vs. domestic concerns.

We can describe the transmission of monetary policy and how the changes in the policy rate settings flow through to economic activity and inflation. This is a complicated process with a high degree of uncertainty around the timing and size of impact on the economy. At a high-level, five transmission channels are adapted to influence markets and the real economy. The transmissions effects are demonstrated

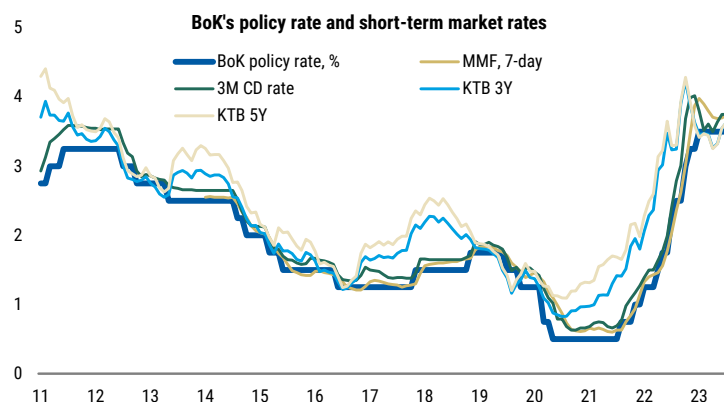
through:

- (1) **interest rates:** through short-term and long-term market rates and deposit and lending rates of domestic banks. Changes in bank rates extend to aggregate demand via changes in consumption and investment behavior of local participants.
- (2) **asset prices:** impact extends to local real and financial asset prices including real estate, stocks and bonds. The changes in the asset and level of wealth of households and corporates dictate consumption and investment decisions.
- (3) **money & credit:** local banks' lending behaviors are immediately affected as they re-assess their lending and liquidity ratios, redemption capacity of borrowers and reserve level under the regulations. This affects overall borrowing behaviors and flows of funds across domestic households, corporates and FIs.
- (4) **exchange rates:** impact on the value of the domestic currency is highly linked to changes in the policy rate and relative changes in the rate levels of other central banks. Lower/higher interest rates can reduce/increase investor interest in the local/overseas assets, influencing shifts in asset movements.
- (5) **inflation expectations:** on the sentiment front, inflation expectations are largely affected by changes in the policy rate. Both the households and corporates/employers' inflation expectations dictate the overall consumption, goods and services price and wage-setting behaviors. Eventually the impact extends to overall real economic performance.

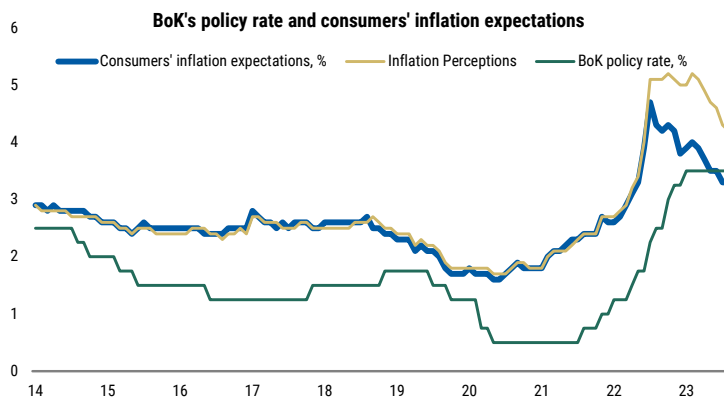
Exhibit 4: Monetary policy transmission mechanism



Source: Source: Bank of Korea, Morgan Stanley Research

Exhibit 5: Policy rate closely influences short-term market rates

Source: Bank of Korea, Morgan Stanley Research

Exhibit 6: BoK policy rate transmits to consumers' inflation expectations

Source: Bank of Korea, Morgan Stanley Research

Monetary Policy Committee and operations

The top policy decision-making body of the BoK is called the Monetary Policy Committee (MPC). The MPC officially meets every first two months of each quarter, holding a total of eight meetings per year. In the case of extraordinary circumstances (i.e., economic shocks, natural disasters, national emergencies), the Committee can call for special meetings in addition to the official schedule. The Governor can make a call to convene on their own motion, and two or more members can also call for a special meeting. The last instance where the Committee called for an emergency policy meeting was in March 2020, when the Committee made a 50bp cut at the onset of the COVID pandemic shock driving extreme volatility in the local financial markets and casting a high level of uncertainty on the real economic outlook.

The MPC's eight regular policy meetings each year take place on Thursdays. Attendance of at least five members out of seven is required for a simple majority decision. During the last month of each quarter, the members attend meetings on Financial Stability, i.e., four times a year. Here, they discuss submitted agenda items and exchange opinions among staff members on various topics across current economic and financial market developments.

The policy meeting starts at 9:00am KST on the day of the decision announcement.

Around 9:20am KST, local media reports on the decision, followed by the release of a formal policy statement on the policy decision at around 9:40-10:00am. Immediately after the meeting, the Governor of the BoK holds a press conference, available in Korean on live public streaming, to keep the markets updated on the latest monetary policy stance of the MPC. The post-meeting conference entails about an hour-long Q&A session to address any questions that might arise regarding that day's decision.

In order to promote transparency and market communications on the monetary decision, the BoK releases the minutes of each meeting to the public on the first Tuesday two weeks after the meeting. The minutes extensively capture the discussions on the various macroeconomic issues, global and local financial markets, as well as inflationary landscape and outlook. The original version in Korean usually offers 30-40 pages of transcript while the abridged English version is also available on the same day on the main website.

Exhibit 7: Key dates for 2023 MPC meetings

#	Policy meetings	Minutes released
1	Friday, 13 Jan	Tuesday, 31 Jan
2	Thursday, 23 Feb	Tuesday, 14 Mar
3	Tuesday, 11 Apr	Tuesday, 2 May
4	Thursday, 25 May	Tuesday, 13 Jun
5	Thursday, 13 Jul	Tuesday, 1 Aug
6	Thursday, 24 Aug	Tuesday, 12 Sep
7	Thursday, 19 Oct	Tuesday, 7 Nov
8	Thursday, 30 Nov	Tuesday, 19 Dec

Source: Bank of Korea, Morgan Stanley Research

Committee members

The BoK's MPC consists of seven members including the Governor and the Senior Deputy Governor. Appointed by the President and approved after the National Assembly's hearing process, the Governor of the BoK chairs the Committee meeting and looks over the monetary policy decisions. He serves a four-year term, with one single consecutive term re-appointment permitted. The Senior Deputy Governor is also appointed by the President upon the Governor's recommendation.

The other five non-BoK members are each nominated by the heads of five different key institutions in Korea before being appointed by the President: Ministry of Economy and Finance (MoEF), Governor of the BoK, Head of the Financial Services Commission (FSC), Head of the Korea Chamber of Commerce and Industry (KCCI) and Head of the Korea Federation of Banks (KFB).

Members serve on a full-time basis during their term in the MPC. In order to prevent overlapping of the service period, the Governor, and nominees from MoEF, KCCI, and KFB serve a four-year term and the Senior Deputy Governor of the BoK and those two members nominated by the Governor of the BoK and the Chairman of the FSC serve a three-year term each.

The MPC currently has a hawkish bias given persistent inflation. Based on publicly available information, We identify three out of the five members excluding the Governor and the Deputy Governor as hawkish, with two members demonstrating an ultra-hawkish

tone (YJ Cho and YK Suh). Amid a stabilizing inflation path, two members continue to show openness to further hikes. Out of the remaining two members, we identify one dovish member (CS Park) who we expect to have a preference for a lower-rate environment in order to support overall economic growth based on his career background at the government body promoting growth. And the last member with a neutral tone based on balanced arguments around rate decisions in the meeting minutes.

Governor of the BoK

The Governor of the BoK is Chang Yong Rhee. Mr Rhee is the first BoK Governor to come from external institutions as opposed to previous governors who have had the majority of their career built within the BoK. As a renowned economist and an experienced policymaker, Mr Rhee brings in a wide range of experience on macroeconomics and policy from international organizations, including the International Monetary Fund (IMF), Asian Development Bank (ADB) and the World Bank.

Starting his term during the pandemic period of April 2022, Rhee has taken a proactive stance towards economic normalization amid strong upward inflationary pressure. Fighting against the elevated level of inflation post-pandemic, Mr Rhee has adopted a hawkish tone so far; the MPC under Mr Rhee has continued the rate hiking cycle which started in August 2021 under the predecessor Juyeol Lee, raising the policy rate from 0.5% to 3.5% in January 2023. The latest policy guidance has been hawkish bias from Mr Rhee, holding the current elevated rate for longer.

Mr Rhee is known to emphasize policy transparency, and openly communicates his stance on the Korean economy and the monetary environment. Mr Rhee has demonstrated a proactive gesture toward market communication at the post-policy meeting press conferences. He also actively participates in the global central bank events – Mr Rhee expressed his thoughts on the BoK and the Fed at the session event at the Jackson Hole meeting in August 2022.

While he has illustrated a hawkish stance so far, we believe Mr Rhee may have a dovish bias on a longer term outlook. Based on his studies and comments, Mr Rhee is known to value private sector-driven economic growth and promotion of structural reform for sustainable economic developments in Korea. Mr Rhee often acknowledges Korea's long-term outlook, and concern around chronically-low growth and the need for structural reform. At a recent business forum held by the KCCI, Mr Rhee emphasized the need for industrial reform and keenly pointed out economic challenges around the rigidity in the domestic labor market, pension and education system. At the post-policy press conference in July 2023, Mr Rhee reiterated the need for structural reform in addition to the need to adjust cyclical policy by the monetary and fiscal policy bodies.

Exhibit 8: Current MPC members as of September 2023

Position	Name	Tenure	Policy stance/ Recent career highlight	Nominating body
Neutral to hawkish				
Governor	Chang Yong Rhee	21 Apr 2022 - 20 Apr 2026	Career highlight: Chief economist, Asian Development Bank Director, Asian and Pacific Department, IMF Education: BA in Economics, Seoul National University PhD in Economics, Harvard University	Appointed by President
Senior Deputy Governor	Sangdai Ryoo	21 Aug 2023 - 20 Aug 2026	Executive Vice President, Korea Housing Finance Corporation Deputy Governor, BoK Head, Fixed Income Markets, BoK BA in Economics, Seoul National University PhD in Economics, State University of NY	Appointed by President
Ultra hawkish				
Member	Yoon Je Cho	21 Apr 2020 - 20 Apr 2024	Ambassador to Korean Embassy in the US Professor, Chair of Graduate School of International Studies, Sogang University BA in Economics, Seoul National University MA and PhD in Economics, Stanford University	MoEF
Ultra hawkish				
Member	Young Kyung Suh	21 Apr 2020 - 20 Apr 2024	Director General, Sustainable Growth Initiative, Korea Chamber of Commerce and Industry Deputy Governor for Research, BoK BA in Economics, Seoul National University MA and PhD in Economics, George Washington University	KCCI
Neutral				
Member	Sung Hwan Shin	28 Jul 2022 - 12 May 2026	President, Korea Institute of Finance Member of Steering Committee, KIC BA in Economics, Seoul National University MA and PhD in Management and Finance, MIT	KFB
Dovish				
Member	Chunsup Park	21 Apr 2023 - 20 Apr 2027	Deputy Minister, Budget Office, MoEF Administrator, Public Procurement Service BA in Int. Trade, Seoul National University MA in Economics, University of Manchester	FSC
Hawkish				
Member	Yongsung Chang	21 Apr 2023 - 20 Apr 2027	Chairman, Macroeconomics Division, National Economic Advisory Council Professor of Economics, Seoul National University BA and MA in Economics, Seoul National University PhD in Economics, University of Rochester	BoK Governor

Source: Bank of Korea, Morgan Stanley Research

*Note: Policy stance of each member is our assumption based on historical voting pattern, comments to the media, previous academic works

Latest monetary policy trend and our outlook

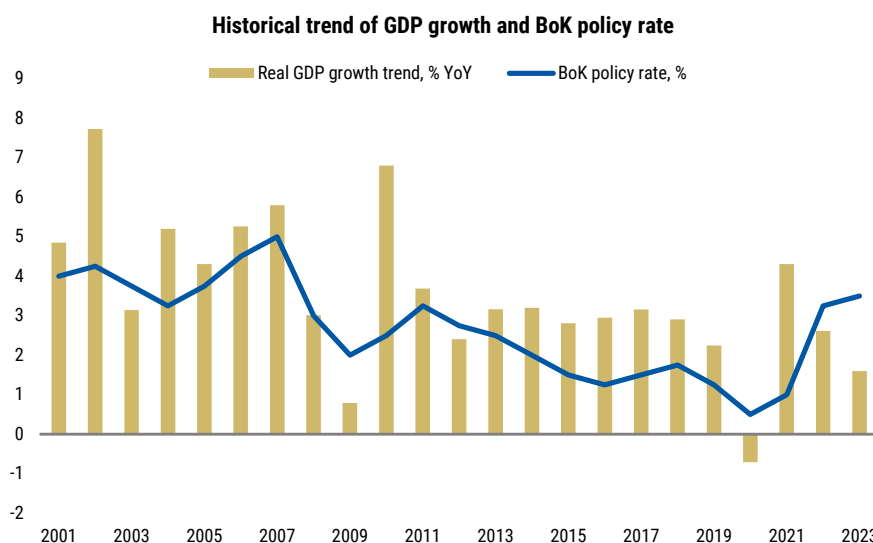
The latest developments of the Korean economy point to an elevated rate

environment for longer. Since the economic re-opening during the summer of 2022, the domestic economy has faced a strong rebound in private consumption driven by pent-up demand on services and recovery in the labor market. Of the headline GDP growth of 2.6% in 2022, private consumption contributed 1.8pp, mostly led by strength in the

household spending. This has driven strength in demand-pull pressure, leaving the domestic CPI inflation at 4.0%Y during the first half of 2023 vs. BoK's target of 2.0%. Persisting demand-side inflationary pressure, especially from personal services, will likely mean core inflation takes longer to stabilize. Consumers' inflation expectations also appear slow to stabilize downward. Until headline CPI reaches its target at 2.0% on a medium-term trend, the BoK is expected to keep the current policy rate at 3.5% to tame inflation further.

Historically, the BoK has carefully conducted monetary policy in tandem with the outlook on the economy's overall price level, which is closely linked to overall growth activity. In the pre-pandemic period, the BoK for nearly a decade adopted a gradual easing stance as the overall GDP growth trend remained weak between 2-3% (2011-19). During this period the BoK tightly managed the policy rate in a range-bound trend between 1.75%-3.25%, avoiding a dramatic spillover impact onto the economy. When the COVID pandemic hit the economy, the BoK swiftly moved to a rapid easing cycle, cutting the policy rate to the historical low level of 0.5% from 1.75% in 2020.

Exhibit 9: BoK has carefully conducted policy to promote stable growth



Source: Bank of Korea, Morgan Stanley Research

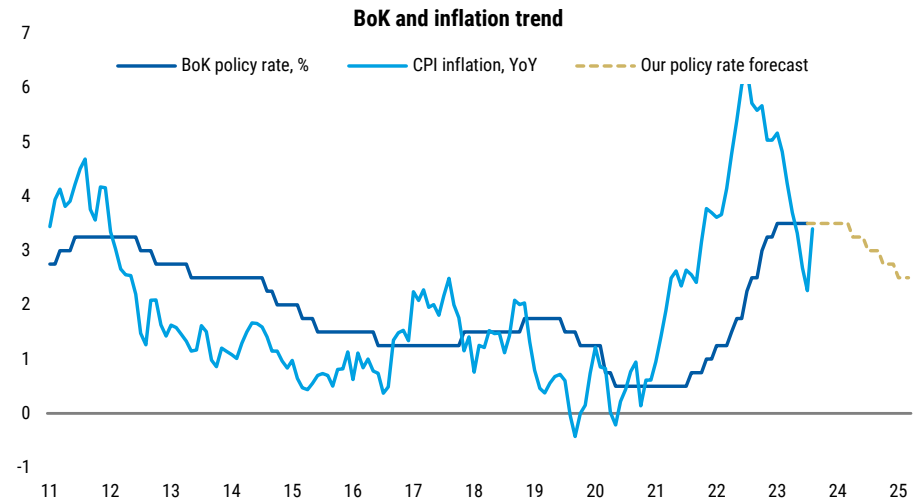
After providing a widely accommodative environment with low rates during the pandemic, the BoK was one of the first major central banks to start normalization. Driven by the fast accumulation of domestic household debt, the BoK started to withdraw from an easy monetary stance from August 2021. Later driven by strong inflationary pressure and the global rate hiking cycle, the BoK has normalized the policy rate during the past 1.5 years to 3.5%, making 300bp of rate hikes. Reflecting a strong rebound in growth and accelerating price trend amid the commodities price spike and demand strength in the economy, the BoK's policy rate at 3.5% as of August 2023 points to a tight monetary policy backdrop.

Most recently, the BoK has showed increased concern around financial imbalance vs. inflation. From July this year, the tone of MPC members has turned even more hawkish on re-accelerating household debt growth while inflation has been downwardly stabilizing. The domestic growth slowdown is also receiving more attention, but is not likely to be an immediate concern, at least not until towards year-end when we forecast consumption to

deteriorate more notably.

As we look forward, we expect the growth recovery on the export front will be gradual while domestic consumption slows further, as we believe pent-up demand likely has been fulfilled after a year since the re-opening. Amid inflation stabilizing downward for rest of the year, the BoK will also closely monitor the recent rebound in the housing market sentiment, which led to a strong increase in mortgage loan growth.

Exhibit 10: Our outlook on the policy rate



Source: Bank of Korea, Morgan Stanley Research estimates

Lastly, the BoK is likely to maintain a more conservative stance overall due to rising downside risk to growth in the coming years. In terms of policy rate forecast, we maintain our view of no rate change in the rest of this year, with the first rate cut in 2Q24. Based on the increased hawkishness around financial stability concerns and lingering uncertainty around the inflation trend, we expect the BoK to continue the current tight monetary stance. However, we expect peak pressure on households from the elevated rate level in 4Q23, after we expect pent-up demand to dissipate. A more visible consumption slowdown from year-end could nudge the MPC to turn more dovish next year.

Fiscal Policy Framework

Kathleen Oh

Ministry of Economy and Finance

The central government's fiscal stance and outlook on the medium-term growth path are critical to overall economic performance as well as financial market activity. By impacting aggregate demand and output via conducting fiscal policies and making budget allocations, the government's economic policy direction and goals for national development greatly influence the behaviors of economic participants and financial market players, both on the domestic and external fronts.

The role of central government is taken by the Ministry of Economy and Finance (MoEF). It is the top economic policy-making body, established in 1948. It serves a critical role across various layers of the economy. To name a few, the Ministry implements the overall direction of the mid- to long-term economic development plan, allocates fiscal resources, monitors budget implementation, manages financial markets and FX policies, and coordinates economic and fiscal policies with key institutions such as the BoK. It is headed by the Deputy Prime Minister, who serves a dual position as the Minister of Economy and Finance, and by two Vice Ministers. A total of 15 bureaus are under the MoEF's oversight.

Exhibit 11: Organization of MoEF

Deputy Prime Minister and Minister of Economy and Finance	
Two Vice Ministers	
11 Bureaus	4 Government agencies
Economic Policy Bureau	Budget Office
Policy Coordination Bureau	Planning and Coordination Office
Economic Structural Reform Bureau	Korea Lottery Commission
Future and Social Economy Bureau	Tax and Customs Office
International Finance Bureau	
International Economic Affairs Bureau	
Development Finance Bureau	
Treasury Bureau	
Fiscal Innovation Bureau	
Fiscal Management Bureau	
Public Institutions Policy Bureau	

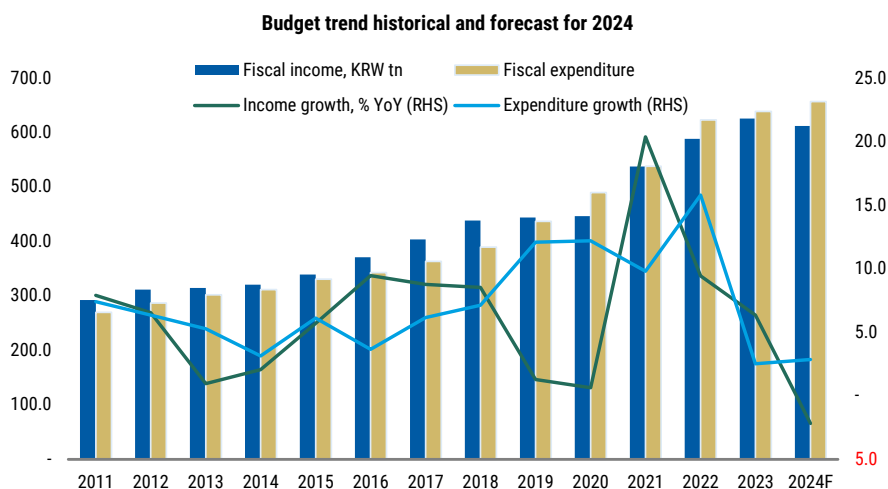
Source: MoEF, Morgan Stanley Research

Fiscal position

Korea currently runs fiscal deficits. Historically, the consolidated fiscal balance recorded surpluses averaging 1% of GDP since the Global Financial Crisis. The consolidated fiscal balance refers to the fiscal balance including social security funds (SSF). But from 2019 when government expanded its agenda and with the fiscal activism during COVID, this fiscal balance started to deteriorate notably. As of end-2022, the consolidated fiscal balance recorded a total deficit of KRW64.6 trillion, or 3.3% of GDP, and the managed fiscal balance (excluding social security funds, but including the national pension fund, teachers' funds, employment insurance fund, and the industrial accident compensation

insurance) recorded a deficit of KRW117 trillion, 5.9% of GDP.

Exhibit 12: Annual budget expenditure growth slowed since 2022 due to falling income

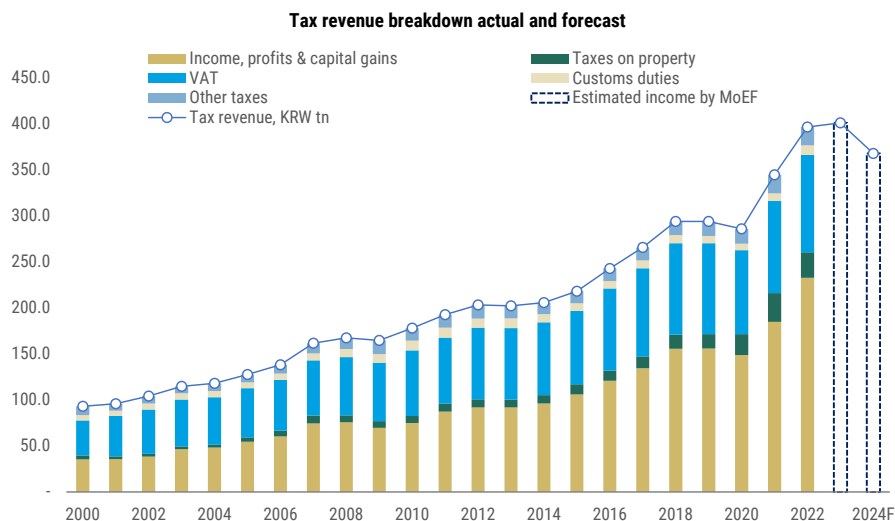


Source: MoEF forecast, Morgan Stanley Research

In terms of fiscal income sources, tax revenue is the main resource. Fiscal dependency on tax collection remains the largest at 80% of total fiscal revenue vs. non-tax sources. Over the past 20 years since 2000, annual average tax revenue in Korea has been KRW200 trillion (2000-22), with income and capital gains tax accounting for the largest share of 48%, followed by 36% contributed by value-add taxes.

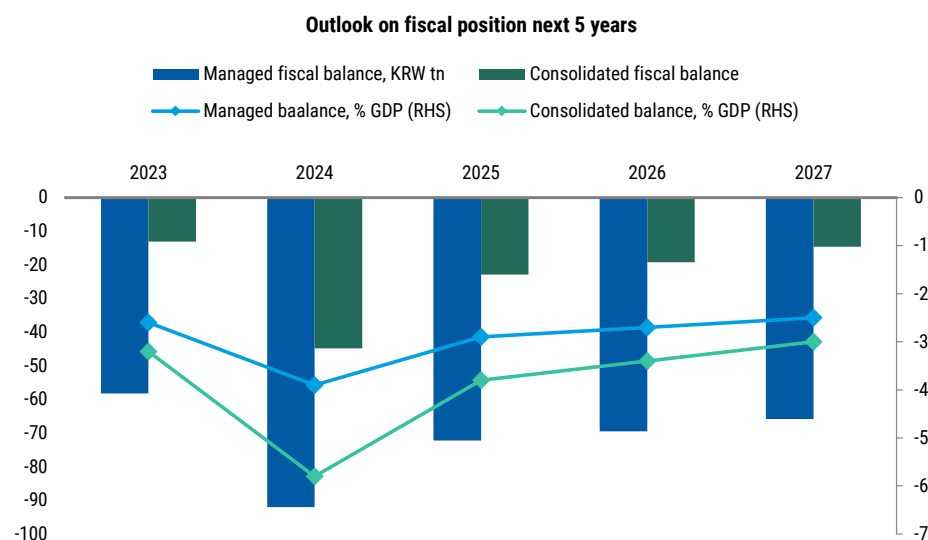
In view of the shortfall in tax receipts as of mid-2023, we expect the deficit trend in the fiscal balance to continue. As of June 2023, tax revenue came in KRW282 trillion vs. realized expenditure of KRW294 trillion, which resulted in a consolidated fiscal deficit of KRW55.4 trillion. The managed fiscal balance also recorded a deficit of KRW82.9 trillion. Given the faster pace of shortfall and dimming outlook on sources of tax receipts amid slower economic performance this year versus last year, we expect a deteriorating fiscal position in 2024 before improvement from 2025 onwards.

Exhibit 13: Outlook on tax receipts remains soft; expect fiscal deficits to continue



Source: MoEF forecast, Morgan Stanley Research

Exhibit 14: Medium-term fiscal outlook guides for narrowing of fiscal deficits in the next 5 years



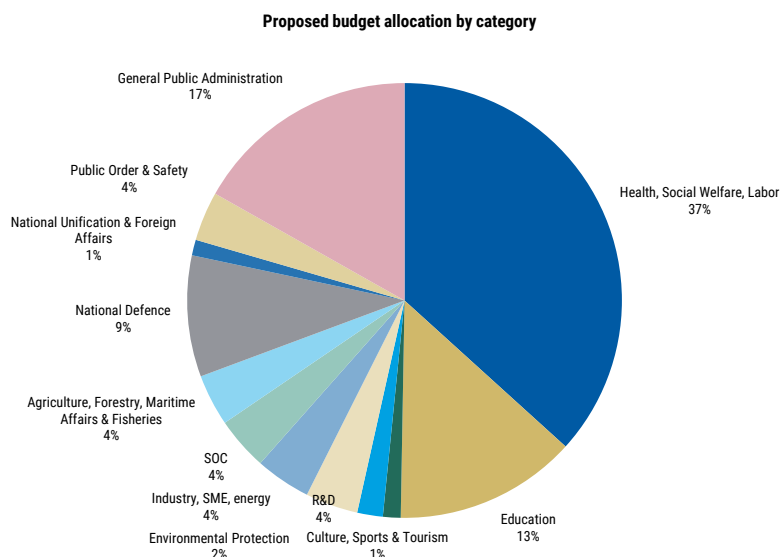
Source: MoEF forecast, Morgan Stanley Research

Government budget

The Korean government revealed the latest draft of its budget for 2024 on 29 August 2023. Historically, conservative governments have pursued fiscal prudence, which we note has continued in the latest budget draft with little increase in spending. After the submission of the draft for National Assembly review on 1 September, Parliament is due to approve the budget during the first week of December, before implementation starts in January.

Historically, the government puts heavy focus on social welfare which has the largest share of allocation. From around 25% of the total budget back in 2010, spending on social welfare rose to >30% by 2018 onwards. The boost in welfare reflects support for the household sector. This year again, the allocation plan outlined the government's agenda on supporting the household sector recovery: while the size of the headline budget increase was limited, the government continued to put major emphasis on social welfare and the labor market, in a way continuing the policy philosophy of "big government" in the household sector.

Following welfare, the second largest allocation goes to general public administration. In the past two decades, spending on public administration has remained unchanged at between 15-17% of the total budget. Education takes the third largest share with 12-15% of total spending, followed by national defence with a share of around 9-10%. The remaining eight sectors received less than 5% each by ministry.

Exhibit 15: Budget proposal on spending allocation by ministries for 2024

Source: MoEF, Morgan Stanley Research

Government debt

Korea maintains a sound level of government debt in proportion to the economy.

'Government debt' is defined as the sum of gross central and local government debt in Korea. As of 2022, Korea's government debt-to-GDP came to 49.7%, far below the OECD average of 110% and below the OECD's recommended ratio of 60%. Historically the Korean government has illustrated a conservative bias towards government spending in the long-term outlook given an aging population where they forecast an increase in government debt. Fiscal activism only took place during times of economic crisis and at times of unexpected economic challenges, such as the COVID pandemic. The government debt-to-GDP jumped to 44% in 2020 and 49% by 2022 from 37% in 2019.

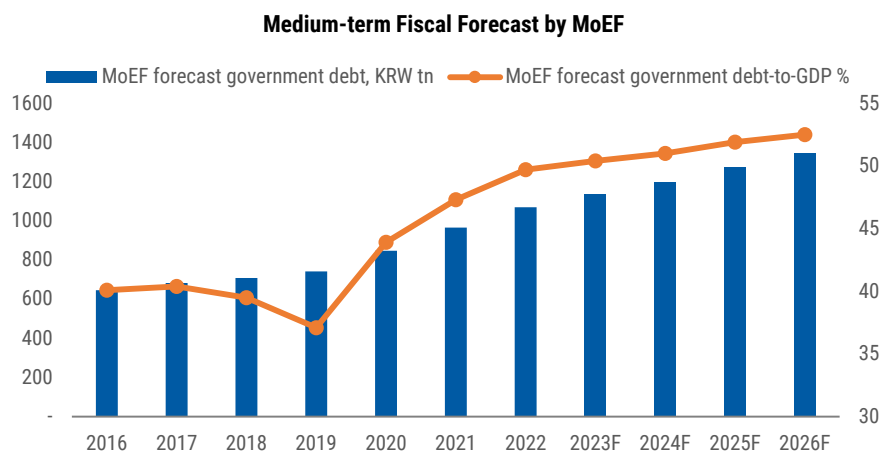
Looking at the government's medium-term outlook for the next four years, the conservative approach looks like it will continue to minimize debt growth. The MoEF forecasts debt-to-GDP to reach around 52.2% by 2026 versus 48.8% in 2023. The government plans to contain debt-to-GDP growth by forecasting its increase by 3pp in the remainder of current government under President Yoon versus a 10pp increase before the pandemic.

Nonetheless, the widening shortfall in the fiscal position looks likely to increase the risk of faster accumulation of government debt in the near term. As fiscal deficits are financed through borrowing from the BoK and issuing Korea Treasury Bonds (KTBs), the weak outlook on tax receipts based on a sluggish economic recovery guides towards a faster rise of debt. While short-term borrowing from the BoK (usually <1 year) remains a minimal share, the bulk of financing comes from newly issuing KTBs. As of 2022, the size of treasuries within government debt reached KRW1,035 trillion, 97% of the total level (KRW1,068 trillion).

We note the latest IMF report pointing out Korea's fast increasing government debt-to-GDP ratio. According to the IMF's classification, Korea's national debt is increasing faster than those of other non-reserve currency peers. In 2021, the average of its peers stood at

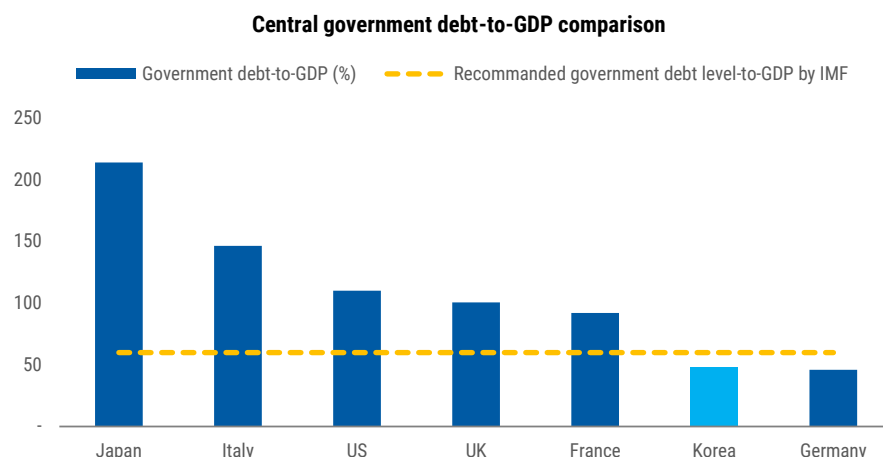
55.6% versus Korea at 51.3% in IMF's estimate. They forecast Korea's level has jumped to 54.4% in 2022, and recently adjusted the estimate to 55.3%.

Exhibit 16: Medium-term outlook guides for a relatively contained pace of increase



Source: MoEF, Morgan Stanley Research (F) forecast

Exhibit 17: Central bank debt-to-GDP comparison shows Korea's healthy government debt level (as of 2022)



Source: IMF, Morgan Stanley Research

Supplementary budget

At times of unexpected economic challenges or events, the MoEF assists financing with a supplementary revised budget. While the current government aims to stay prudent and looks to tightly manage government debt, Congress may come to an agreement to approve a supplementary budget, also known as an extra budget on top of the original budget that can be allocated in the middle of the current fiscal year. Article 89 of Korea's National Finance Act delineates the occasions when the government can formulate the extra budget bill. It states: "The Government may formulate any supplementary revised budget Bill, if one of the following events occurs, thereby making it necessary to revise the budget already finalized:

- A war or large-scale disaster (referring to damage caused by natural disasters and

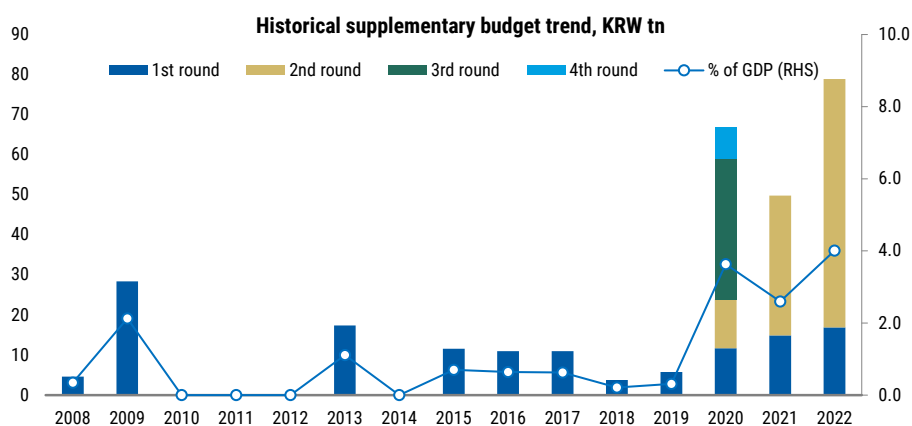
social disasters defined in Article 3 of the Framework Act on the Management of Disasters and Safety) breaks out;

- *A significant change in circumstances at home and abroad, such as an economic recession, mass unemployment, change in inter-Korean relations or economic cooperation, occurs or is likely to occur;*
- *The expenditure the State is obligated to pay pursuant to statutes is incurred or increased."*

Historically, Korea executed small-size extra budgets prior to the Global Financial Crisis of 0.4% of GDP, with the exception of the Asian Financial Crisis (1.7% in 1997) and credit card crisis (2.1% in 2002). More recently, however, the frequency of supplementary budget execution has become an annual event. When there was a need to provide additional funding during the 2020-22 COVID pandemic, the size jumped to close to 4% of GDP.

In the process of executing a supplementary budget, the government seeks funding through the issuance of KTBs, at times of scarce funding available from other means. According to Article 90 of the National Finance Act, the government is to seek funding for the extra budget in the order of: 1) tax revenue from the previous tax year; 2) available tax revenue from the current tax year; 3) income from equity transactions of state-owned entities; 4) special reserves at the BoK; and 5) lastly through issuance of new treasuries.

Exhibit 18: Korea has executed sizable supplementary budgets during the pandemic



Source: MoEF, Morgan Stanley Research

Fiscal Rule

The government seeks to sustain sound fiscal conditions through the implementation of the Fiscal Rule. The Korean government first proposed the bill for Fiscal Rule in October 2020, citing the criticality of tightly managing government debt growth, which could have threatened its sovereign rating. Korea currently maintains a credit rating of Aa2 by Moody's, AA by S&P and AA- by Fitch, with a stable outlook. This comes against the backdrop of rising government debt and an outlook of increased government expenditure amid structural changes, namely a rapidly aging population.

The Fiscal Rule consists of a central government debt-to-GDP capped at 60% and a floor on the consolidated fiscal balance controlled at a deficit of no larger than 3% of GDP. It is aimed to operate on 'flexible targets' where, in case the debt level reaches 60% of GDP,

the floor level tightens to a permitted deficit equivalent of 2% of GDP. This is described as 'debt brake' mechanism, where the deficit floor automatically tightens when the debt-to-GDP ratio rises. The only exceptions to the rule would be in the case of war, national disasters or an economic recession.

Korea's proposed Fiscal Rule formula:

$$\circ \text{ [Government debt / 60\%] * [Consolidated Fiscal Balance / (-3\%)]} \leq 1$$

Although a swift enactment looks less likely, continued dialogue around implementation of the Rule captures Korea's strong commitment to promote prudent fiscal policies. Currently, the government employs the rule as soft guidance, as the bill is conditional on parliamentary approval. The previous government proposed a 3-year grace period during the pandemic. The current government targets implementation from 2025 onwards and plans to revisit for revisions on necessary basis every five years.

Bond Market

Gek Teng Khoo, Min Dai

Overview of Korea's domestic bond market

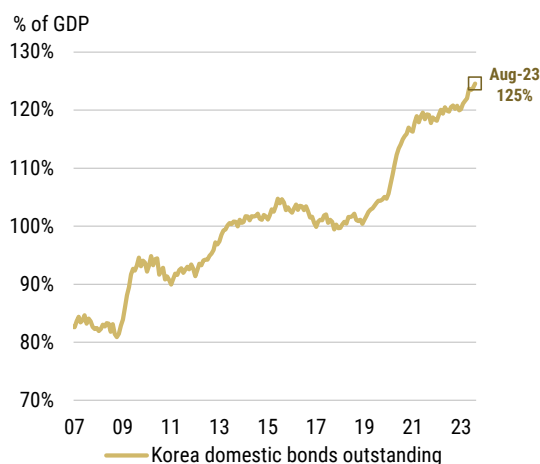
A US\$2 trillion market: The size of Korea's bond market has increased steadily over the years, with the pace accelerating during the Covid pandemic amid the ramp up in fiscal spending. Currently, the total amount of all bonds outstanding stands at ~KRW 2,709 trillion, or ~US\$2 trillion ([Exhibit 19](#)).

There are eight main types of bonds in the Korea market:

1. **Government bonds:** issued by the central government
2. **Municipal bonds:** issued by local governments
3. **Special bonds:** issued by state-owned corporations established under special laws
4. **Monetary stabilisation bonds (MSBs):** issued by the Bank of Korea to manage liquidity
5. **Corporate bonds:** issued by non-financial companies
6. **Bank bonds:** debentures issued by banks, including the three policy banks
7. **Other financial bonds:** issued by non-bank financial companies
8. **Asset-backed securities (ABS):** bonds that are collateralised by a pool of underlying assets

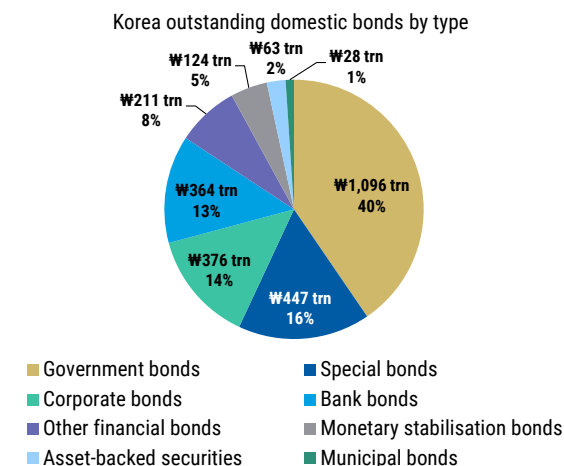
The government bond market is the largest out of the eight different categories of bonds. With a size of KRW 1,096 trillion (~US\$825 billion), it accounts for 40% of the overall bond market ([Exhibit 20](#)). The special bonds market is the next largest, at KRW 447 trillion (~US\$337 billion) or 16% of the overall bond market, followed by the corporate bond market (~KRW 376 trillion or US\$283 billion).

Exhibit 19: Korea's bond market size has grown to ~125% of GDP



Source: KOFIA, Haver Analytics, Morgan Stanley Research

Exhibit 20: The government bond market is the largest

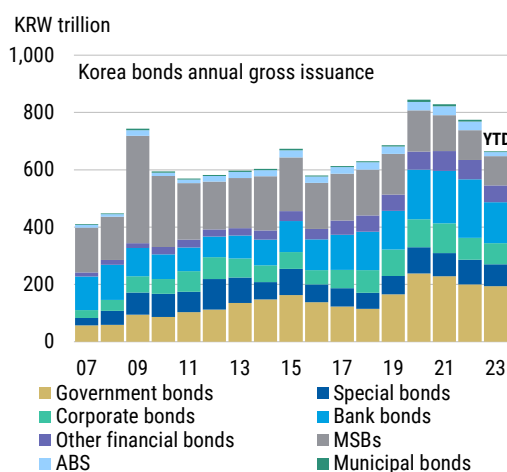


Source: KOFIA, Haver Analytics, Morgan Stanley Research

Government bonds, bank bonds and MSBs are most actively issued: The size of each bond market is not always reflective of how actively it is issued. For example, with a size of just KRW 124 trillion (~US\$94 billion), MSBs may not seem like a large market as it accounts for just 5% of the overall domestic bonds outstanding. But this is mainly due to their shorter maturity profile – MSBs are issued with a maximum maturity of 3 years.

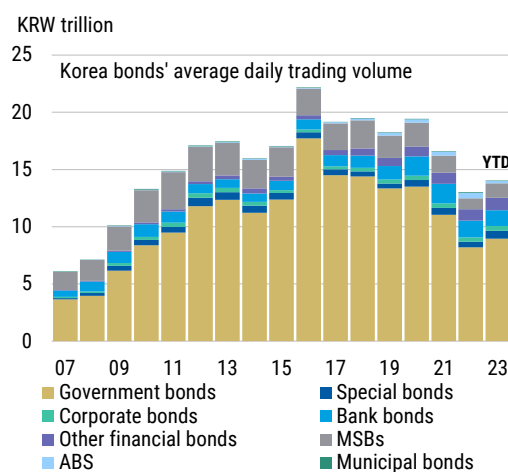
MSBs are in fact one of the most actively issued bonds, along with government bonds and bank bonds ([Exhibit 21](#)). Before 2019, the issuance volume of MSBs had even exceeded that of government bonds each year given the more conservative fiscal stance.

Exhibit 21: Bank bonds, government bonds and MSBs see the largest issuance



Source: KOFIA, Haver Analytics, Morgan Stanley Research

Exhibit 22: Average daily trading volumes in Korea bonds have been declining



Source: KOFIA, Morgan Stanley Research

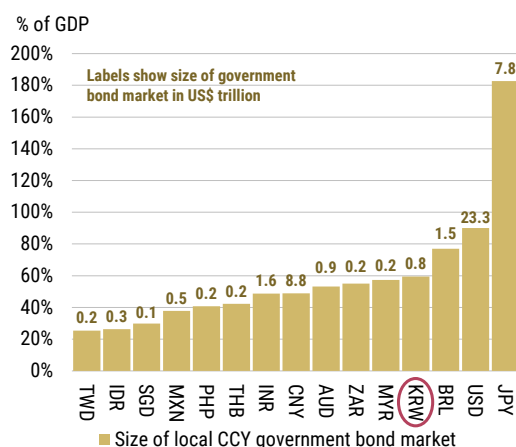
Trading volumes have declined: The trading volume of Korean bonds is correlated with price performance. Trading volumes peaked at an average of KRW 22 trillion/day in 2016 alongside the top in bond prices. In the last two years as bond prices have sold off, the trading volume of Korean bonds has declined meaningfully. As bond prices bottomed in October 2022, the trading volume of Korean bonds has started picking up this year to average KRW 14 trillion/day.

Government bonds are by far the most actively traded bonds in the Korea market. With an average daily trading volume of KRW 9 trillion year-to-date, they account for 64% of the total trading volume of all bonds ([Exhibit 22](#)). Bank bonds, MSBs and other financial bonds are actively traded as well, accounting for nearly 10%, 9% and 8% of total bond trading volumes, respectively, this year.

Types of Korean government bonds

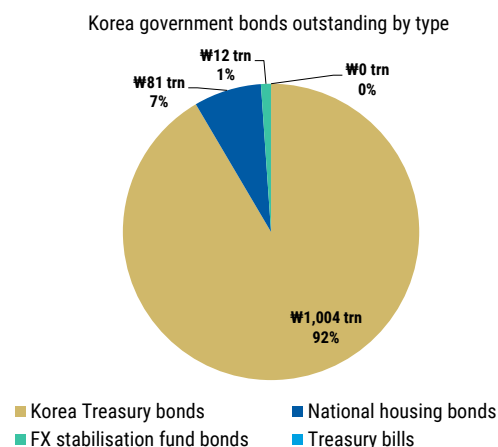
Korea's government bond market is the largest within AXJ when compared to the size of the economy, at 60% of GDP ([Exhibit 23](#)). In absolute USD terms, it is the third-largest government bond market in EM Asia, behind only China and India.

Exhibit 23: Korea is the largest government bond market within AXJ as % of GDP



Source: BIS, Macrobond, Morgan Stanley Research

Exhibit 24: KTBs are the main type of government bonds issued



Source: MoEF, Haver Analytics, Morgan Stanley Research

There are four key types of government bonds issued:

1. **Korea Treasury bonds (KTBs):** issued to finance government spending
2. **Treasury bills (T-bills):** issued to cover temporary shortfalls in the government budget
3. **FX stabilisation fund bonds (FX SFBs):** issued to create a favourable environment for the private sector to issue foreign currency bonds and has the effect of raising FX reserves
4. **National housing bonds (NHBs) Type 1:** issued to raise funds for housing construction projects

KTBs are the main type of government bonds issued, accounting for 91% of the total government bond market with a size of KRW 1,004 trillion (US\$757 billion). This is followed by NHBs (7% of the market) and FX SFBs (Exhibit 24).

Within KTBs, there are two types of bonds issued:

1. **Nominal KTBs:** pay a fixed coupon rate based on a fixed principal amount
2. **Inflation-linked KTBs (KTBis):** pay the principal and coupons based on changes in CPI over time

Exhibit 25: An overview of the types of Korean government bonds

Type of government bond	Purpose of issuance	Maturity	Issuance method	Coupon rate
Korea Treasury Bonds (KTBs)	Finance government spending	2, 3, 5, 10, 20, 30, 50 years	Competitive auction	Set at auction
Treasury Bill (T-bills)	Cover temporary shortfalls in budget	Less than 1y	Competitive auction	0%
Foreign Exchange Stabilisation Fund Bonds (FX SFBs)	Create favourable environment for private sector to issue foreign currency bonds	Set at issue	Competitive auction	Set at issue
National Housing Bonds (NHBs) Type 1	Raise funds for housing construction projects	5y	Mandatory placement	1.30%

Source: MoEF, Morgan Stanley Research

Supply: Issuance tenors and timing

Annual KTB issuance plan unveiled in December: The Ministry of Economy and Finance (MoEF) usually makes its budget proposal in August and announces its KTB issuance plan for the year ahead in December. The plan specifies the amount of KTBs it aims to issue on both a gross and net basis, whether it intends to issue more in the front end, belly or long end of the curve, and the amount it plans to issue in the first half versus second half of the year.

Exhibit 26: Actual and net KTB issuance in recent years

KTB issuance (KRW trn)	2016	2017	2018	2019	2020	2021	2022	2023 (planned)	2024 (proposed)
Gross issuance	101.1	100.8	97.4	101.7	174.5	180.5	168.6	167.8	158.8
Redemption	69.3	71	77.1	57.2	59.2	59.9	71.4	106.3	108.5
Net issuance	31.8	29.8	20.3	44.5	115.3	120.6	97.2	61.5	50.3

Source: MoEF, Morgan Stanley Research

Nominal KTBs are issued regularly with seven different maturities: 2-year, 3-year, 5-year, 10-year, 20-year, 30-year and 50-year. Most of the nominal KTBs outstanding are of 10-year and 30-year maturities ([Exhibit 27](#)). In the secondary market, **3-year KTBs are the most actively traded** (~KRW 1.6 trillion/day on average year-to-date). 5-year, 10-year, and 30-year KTBs also see relatively higher trading volumes of ~KRW 600-700 billion/day on average.

Meanwhile, **inflation-linked KTBs are issued only with a 10-year maturity**. The size of the KTBi market is small, at only ~KRW 12 trillion, or ~1% of the total KTB market. Consequently, the trading volume of 10y KTBi is also relatively low at just an average of ~KRW 27 billion/day.

In recent years, the government has been increasing long-end (30-year and 50-year) KTB issuance given the low interest rate environment and strong demand from long-term investors such as insurance companies and pension funds.

Since 30-year bonds were first introduced in September 2012, the share of 30-year bond issuance has jumped from 2% of total annual issuance to 28% of total annual issuance ([Exhibit 28](#)). This has come mainly at the expense of reduced issuance in the belly of the curve, with 5-year KTB issuance falling to just 16% of total annual issuance from 31% in 2012. In October 2016, the government started issuing 50-year KTBs.

Exhibit 27: Most nominal KTBs outstanding are of 10-30y maturities

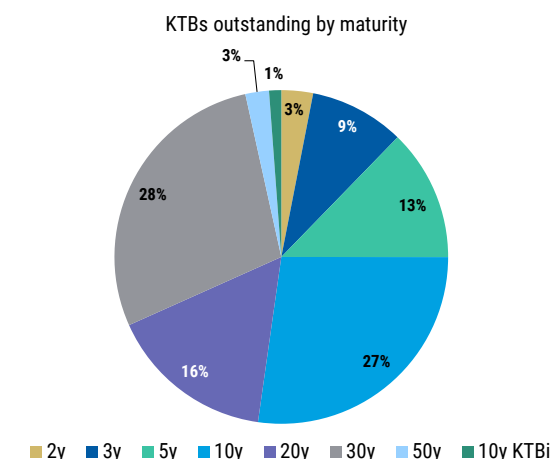
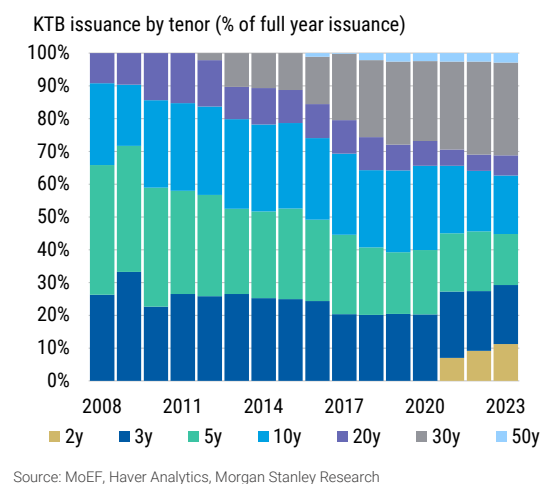


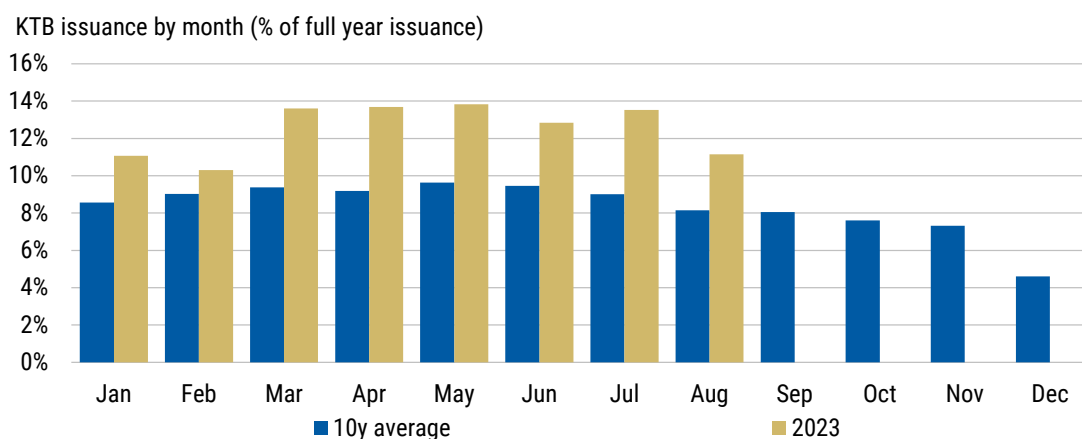
Exhibit 28: The MoEF has been issuing more long-term bonds



Issuance tends to be front-loaded in 1H: In terms of the issuance timing, the government spreads the KTB issuance out over the course of the year while taking into consideration the timing of tax revenue inflows and expenditure needs as well as market conditions.

Similar to several other EM economies, KTB issuance has historically been more front-loaded in the first half of the year, with March, May and June being particularly busy ([Exhibit 29](#)). Issuance then tapers off gradually in the second half of the year, with December issuance being the lowest amid thin market liquidity.

Exhibit 29: KTB issuance tends to be front-loaded in 1H



Supply: How do KTB auctions work?

Auction timing: KTBs are issued as coupon bonds and pay interest every six months. KTB auctions are held on a regular basis to maximise predictability. [Exhibit 30](#) shows the typical auction timing for the different KTB tenors. Nominal KTBs at each tenor are issued every month, while KTBis are issued only every two months, on the first Friday of even months.

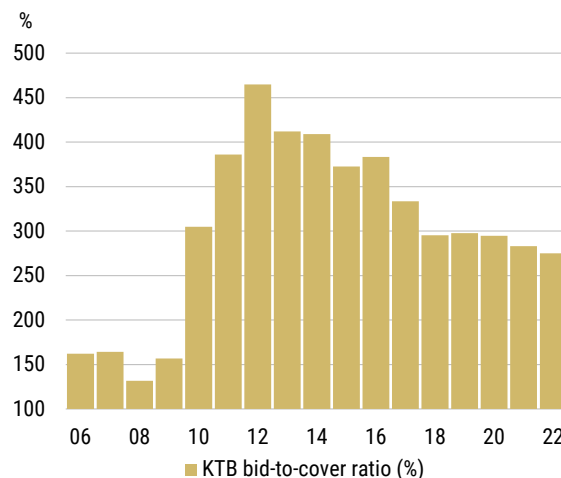
The MoEF releases the issuance calendar for the month ahead in advance, usually by the last Thursday of each month. The release details the auction dates, issuance volumes, buybacks and conversion offers etc.

Exhibit 30: Typical auction calendar for KTBs

Tenor	Typical auction timing
<i>Nominal KTBs</i>	
2-year	Second Tuesday of every month
3-year	Second Monday of every month
5-year	Fourth Monday of every month
10-year	Third Monday of every month
20-year	Fourth Tuesday of every month
30-year	First Monday of every month
50-year	Second Friday of every month
<i>Inflation-linked KTBs</i>	
10-year	First Friday of every other month (even months)

Source: MoEF, Morgan Stanley Research

Exhibit 31: KTBs bid-to-cover ratio has declined



Source: MoEF, Haver Analytics, Morgan Stanley Research

Auction method – Competitive auctions: KTBs are usually issued through competitive auctions, with non-competitive bidding options available too. Most KTB issues are sold to primary dealers via competitive auctions. Currently, the Dutch auction method is being used, whereby bids are ordered in descending order and the lowest price needed to absorb the full issuance amount becomes the price for all successful bidders.

Prior to this, from 2009 to early 2021, the government had been using the differential price auction method, where all bid yields are divided into groups of 3-5bp intervals in descending order and the accepted yield is the highest bid yield in each group.

However, as KTB issuance and bond yields rose significantly during the Covid pandemic, the government switched to the Dutch auction method in March 2021 (which was also used from 2000 to 2009) to encourage active bidding in KTB auctions, given that high bids are not penalised under this method.

There are four non-competitive bidding options available:

1. **For retail investors:** Retail investors can buy KTBs from primary dealers by submitting their desired purchase amount at least one day before the competitive auction. They will purchase the KTBs at the highest accepted bid yield at the competitive auction. No more than 20% of the offered amount in the auction (excluding 50y KTBs) will be allocated first.
2. **For primary dealers:** Primary dealers that have underwritten KTBs in the competitive auction can make additional purchases of 10-35% of the accepted bid amount (depending on their market-making performance) within three business days after the competitive auction.
3. **For primary dealers specialising in STRIPS:** KTB STRIPS are usually supplied on the third business day after the competitive auction. Primary dealers that have the right to purchase these STRIPS can buy up to KRW 20 billion.
4. **For the government:** In the event of lacklustre demand, the government has the

option to underwrite additional issuance volume to the amount that falls short of the month's plan at a specified interest rate. The MoEF will announce the issuance plan on the second Thursday of each month if this option needs to be used.

Demand: Key players in the KTB market

Domestic institutional investors are the main participants in the KTB market, [owning](#) 79.4% of the KTB market as of June 2023. Domestic individual investors owned just 1.2% of KTBs, with the rest owned by foreigners.

Banks are the largest holders of KTBs: Looking at the breakdown by investor type, i.e., including both domestic and foreign investors, KTB investors can be broadly categorised into six main groups ([Exhibit 32](#)):

1. Banks: Banks have been the largest holders of KTBs and have been increasing their holdings steadily through the years, consistent with the trend of rising deposits. Their KTB holdings can also be influenced by the regulatory regime, such as the Liquidity Coverage Ratio (LCR) requirements. As of end-2022, banks' KTB holdings stood at KRW 447 trillion, or 46% of total outstanding KTBs.

2. Insurers: Insurers are the second-largest investors of KTBs, buying mainly long-term KTBs with maturity of more than 10 years for asset-liability management (ALM). Their demand may also be affected by changes in regulations or accounting standards (e.g., IFRS 17).

As liquidity concerns increased in September 2022, insurers sold KTBs to secure cash. But aside from this episode, insurers' KTB holdings have generally been on an increasing trend over the years. We expect this to continue in view of tighter ALM. At the end of 2022, insurers held KRW 232 trillion of KTBs (24% of total outstanding), down from the peak of KRW 268 trillion in March 2022.

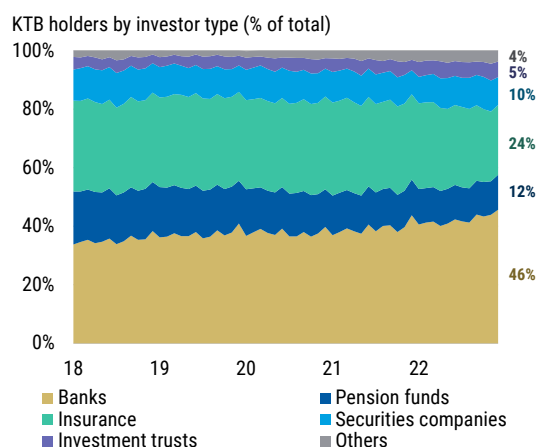
3. Pension funds: Pension funds are the third-largest holders of KTBs, but their share of the market has been declining steadily over the years as the National Pension Service increased its investments abroad. At the end of 2022, pension funds held 12% of outstanding KTBs (KRW 117 trillion), down from 19% at the start of 2014.

4. Securities companies: Securities companies' holdings of KTBs have been increasing at pace with the rate of growth of the KTB market, keeping their share of the market largely stable through the years. Their demand for KTBs is dependent on factors such as the amount of funds under management and market conditions. At the end of 2022, they held 10% of the KTB market (KRW 94 trillion) versus 11% in 2014.

5. Investment trusts: Investment trusts' holdings of KTBs have been on a slight upward trend since 2019 after declining in the prior years. As of end-2022, they held 5% of the KTB market (KRW 52 trillion), unchanged from 2014 but up from the low of 3% in March 2019. Similar to securities companies, their demand for KTBs depends on factors including the amount of funds under management and market conditions.

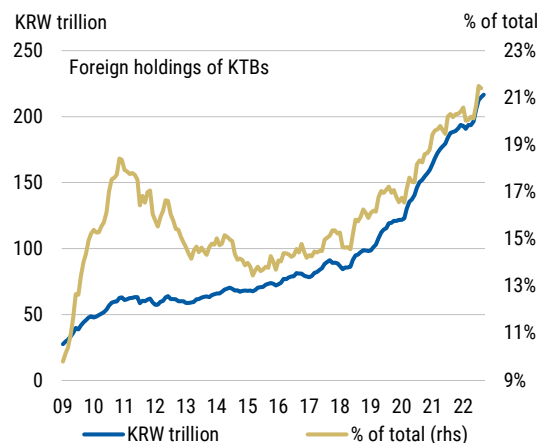
6. Others: Aside from banks, insurers, pension funds, securities companies and investment trusts, other investors hold ~4% of the KTB market (KRW 36 trillion).

Exhibit 32: Banks and insurers are the largest holders of KTBs



Source: MoEF, Haver Analytics, Morgan Stanley Research

Exhibit 33: Foreigners have become larger investors in the KTB market



Source: FSS, MoEF, Haver Analytics, Morgan Stanley Research

Foreign ownership has been rising: Looking at the breakdown between domestic and foreign investors, foreigners had been negligible investors in the KTB market until 2007, due to the lack of an active secondary market. Since then, foreigners have been increasing their KTB holdings, with the pace accelerating from mid-2019.

Currently, foreigners hold KRW 217 trillion of KTBs (as of August 2023), or 21% of the market, doubling from the 10% seen at the end of 2009 (Exhibit 33). According to the MoEF, since August 2017, more than 70% of the foreign demand for KTBs has come from medium- to long-term investors (e.g., global asset managers, foreign central banks and international organisations).

Trading KTBs in the secondary market

In the secondary market, **Korean bonds can either be traded over-the-counter (OTC) or on the exchange market (KRX). Government bonds have usually been traded more actively in the exchange market** compared to the OTC market since 2015, but 2022 was an exception.

OTC market: Trading in the OTC market is done via online messenger, voice or the K-Bond online trading platform. The main participants in the OTC market are institutional investors (e.g., financial institutions and pension funds) and corporates, though individual investors can also participate.

KTBs are the most actively traded in the OTC market (~47% of total trading volume), followed by bank bonds (~15% of total volume) and MSBs (~13% of total volume). There are no restrictions in trading hours, though they generally follow the business hours of financial institutions, i.e., 8:30am-4:30pm KST. Most trades are settled on a T+1 basis, though it can be up to T+30.

Exchange market: Trading in the exchange market is done via the electronic trading system KRX KTB. The main participants in this market are banks and securities firms who are licensed to deal in KTBs. KTBs, MSBs and Korea Deposit Insurance Fund Bonds (DIFBs) are traded on the KRX KTB, but KTBs account for ~98% of all the trading activity. The

trading hours are 9:00am-3:30pm KST, with the trade settlement period being T+1.

Settlement: Trades are settled via the Delivery versus Payment (DVP) method in both the OTC and KRX KTB markets, i.e., the delivery and payment of securities are carried out simultaneously, minimising settlement risk. Under special circumstances, the Free of Payment (FOP) settlement method can be used in the OTC market, whereby the delivery and payment of securities are carried out separately.

In both the OTC and KRX KTB markets, the BoK acts as the payment settlement institution while the Korea Securities Depository (KSD) acts as the securities settlement institution.

Exhibit 34: Summary of OTC versus exchange market

	OTC market	Exchange market (KRX KTB)
Participants	Brokers	Dealers
Trading execution method	Negotiated trading	Competitive bidding
Trading means	K-Bond, Voice	Online bond trading system
Trading unit	Usually KRW 10 billion	KRW 1 billion
Trading hours	Generally 8:30am - 4:30pm KST	9:00am - 3:30pm KST
Settlement period	Mostly T+1	T+1
Settlement method	Delivery versus Payment	Delivery versus Payment
Average daily trading volume	KRW 4.4 trillion	KRW 4.6 trillion

Source: MoEF, KOFIA, Morgan Stanley Research

How do foreigners access the KTB market?

Are there any investment limits? The opening up of Korea's bond market to foreign investors accelerated in December 1997 after the Asian Financial Crisis. Since May 1998, Korea has allowed foreign investment in all bonds and removed the limit on foreign investment in listed bonds. Foreigners can also repatriate their capital investment or interest income without restrictions, though documentation may be required for amounts exceeding US\$100,000 per year.

Investment procedure: Foreigners can access both the OTC and exchange markets. To trade KTBs, they need to first appoint a standing proxy and a custodian who will handle all the necessary procedures for trading securities in Korea and hold custody for the purchased securities. The standing proxy will receive the Investment Registration Certificate (IRC) from the Financial Supervisory Service (FSS).

In order to trade securities, foreigners must open a trading account at a domestic securities company. Foreigners must then open a foreign currency account and a non-resident KRW account at a domestic custodian institution.

Following this, foreign investors can then place orders on their own or via the standing proxy. To settle any trades, foreign investors will deposit funds in foreign currency to the domestic custodian, who will then convert them to KRW and use them to process settlement with the financial intermediaries. [Exhibit 35](#) illustrates how a typical bond transaction by a foreign investor works.

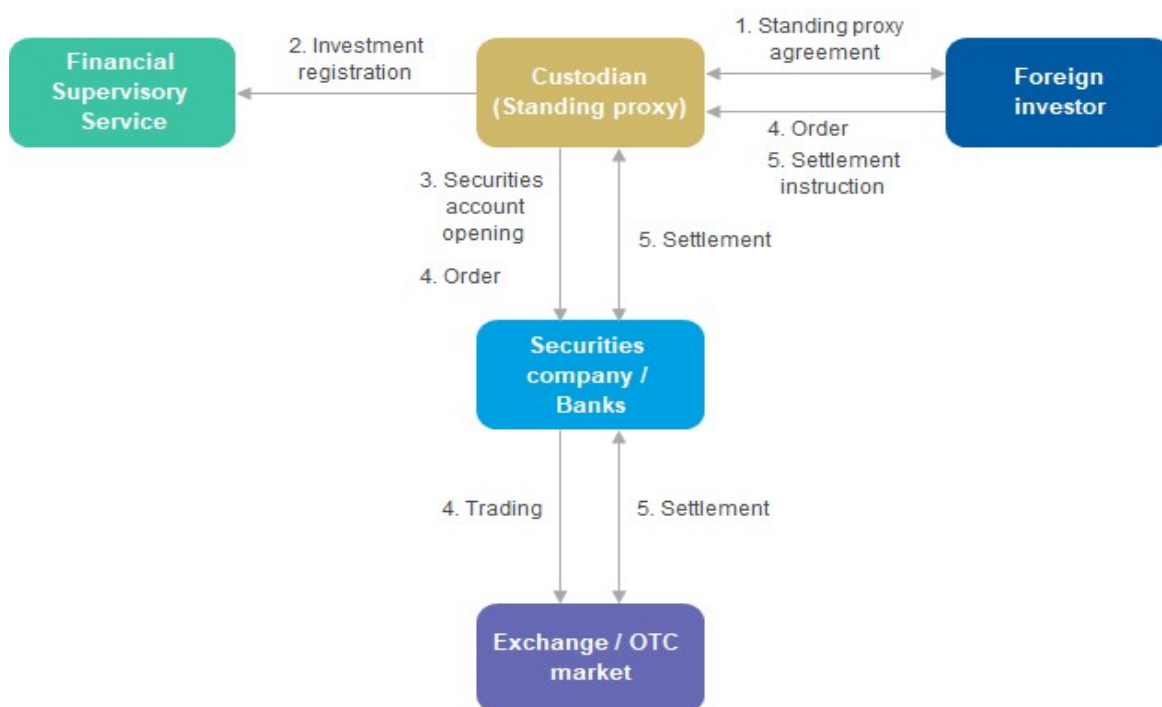
Euroclear in the works: As part of its efforts to get Korea included in the FTSE World Government Bond Index (WGBI), the government has been working with Euroclear to introduce the ICSD-linked omnibus account for government bonds. An agreement has been signed, with the target implementation date currently being 1H24.

With the omnibus account, foreign investors will be able to invest in government bonds through their existing ICSD (International Central Securities Depository) accounts without having to open a domestic trading account, simplifying the process.

Taxation: Previously, foreign investors' interest income and capital gains from their bond investments in Korea were subject to taxation. This was a hurdle for Korea's WGBI inclusion so in late 2022, the parliament passed a bill to exempt taxes on foreign investors' interest income and capital gains from their investments in KTBs and MSBs.

To apply for the tax exemption, foreign investors need to submit [an application form](#) and a certificate of residence to the custodian, overseas investment vehicle or qualified foreign intermediary.

Exhibit 35: Investment procedure for foreigners to buy KTBs



Source: MoEF, Morgan Stanley Research

How can the BoK/authorities influence interest rates?

The BoK has three monetary policy instruments which it can use to manage the level of liquidity in the financial system and influence interest rates: 1) open market operations; 2) lending and deposit facilities; and 3) reserve requirement ratio. Open market operations are the BoK's main tool.

1. Open market operations

There are three key types of open market operations conducted by the BoK: 1) securities transactions; 2) MSB issuance and repurchase; and 3) term deposit facility

(Monetary Stabilisation Account).

Securities transactions refer to the BoK buying or selling bonds to inject or withdraw liquidity from the markets. The BoK can either buy or sell bonds outright in the secondary market (outright transactions), or buy or sell bonds with the condition that it will sell or buy them back after a specified period of time (repo transactions).

Outright transactions are used in a limited way as they affect liquidity permanently and directly affect long-term bond prices and interest rates. Most recently, the BoK bought government bonds in 2020-22 to help stabilise the bond market as issuance ramped up with the Covid-related fiscal spending.

Repo transactions are offered weekly, mostly with 7-day maturities though they can be up to 91 days. When the credit market was affected by heightened concern in late 2022, the BoK turned from selling to buying repos to provide liquidity to the market.

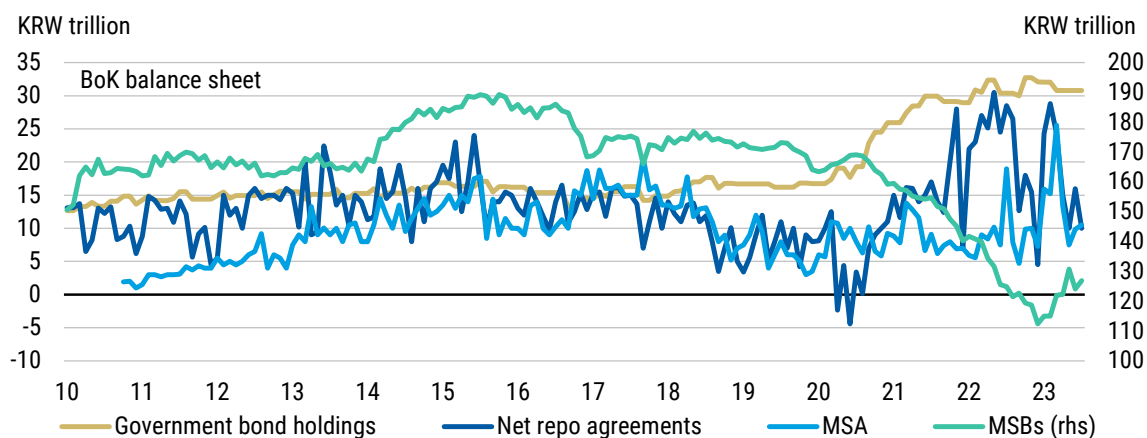
The eligible bonds for securities transactions are usually government bonds and government-guaranteed bonds, but can be expanded by the BoK monetary policy board when needed. For example, the BoK broadened the range of securities eligible for open market operations in April 2020 during the Covid shock, and October 2008 during the GFC.

In April 2020, the BoK also accepted debentures issued by the three policy banks (Korea Development Bank, the Industrial Bank of Korea and the Export-Import Bank of Korea) as well as MBS issued by the Korea Housing Finance Corporation as eligible securities for outright transactions. In addition, it included bonds issued by the Korea Deposit Insurance Corporation for repo transactions.

The BoK also issues bonds (MSBs) regularly: MSBs have longer maturities than repo transactions – they are commonly issued with 3-months to 3-years maturity – so are used for the structural adjustment of market liquidity. See [Overview of MSBs](#) for more details.

The BoK also runs a term deposit facility called the Monetary Stabilisation Account (MSA), where the central bank accepts term deposits from commercial banks to finetune the level of reserves in the banking system and manage unexpected changes. The longest maturity of deposits in the MSA is three months.

Exhibit 36: Amount outstanding of BoK's open market operations



Source: BoK, Haver Analytics, Morgan Stanley Research

2. Lending and deposit facilities

The BoK also has four lending facilities and a deposit facility to supply loans to and receive deposits from banks to manage liquidity. [Exhibit 37](#) shows the different purposes, maturities and interest rates applicable to these facilities.

The BoK currently has KRW 30 trillion of loans outstanding, out of which KRW 28.9 trillion are made under the bank intermediated lending support facility. There are no liquidity adjustment loans outstanding.

Exhibit 37: Overview of BoK's lending and deposit facilities

	Facility	Purpose	Maturity	Interest rate	Collateral
Loans	Liquidity adjustment loans	Provided to banks experiencing fund shortages to limit excess volatility of call rate	1 business day (can be extended up to 3 months)	BoK policy rate + 50bp	- Credit securities acquired by financial institutions through loans - Government bonds, MSBs, Municipal bonds
	Bank intermediated lending support facility	Provide low interest rate funds to banks based on their lending to SMEs, up to a specific ceiling	1 month	0.25 - 2% per annum	- KDB, IBK and KEXIM debentures - MBS issued by Korea Housing Finance Corporation - Special bonds
	Intraday overdrafts	Provided to banks experiencing temporary shortages of settlement funds within a day	Closing hour for funds transfer in the day	- Loan amount < 25% of bank's equity capital: 0% - Loan amount > 25% of bank's equity capital: 3y KTB yield - uncollateralised overnight call rate in last month of preceding quarter	- Select bank debentures - Other bonds issued by public institutions - High-grade (AA- or higher) corporate bonds
	Emergency credit to financial institutions	Provided to banks that are facing deteriorating liquidity or are expected to face pronounced difficulties due to temporary shortage of payment funds	Determined by BoK monetary policy board	Determined by BoK monetary policy board	Determined by BoK monetary policy board
	Special loans	Provided to for-profit enterprises when severe impediments to obtaining funds from financial institutions occur or has a strong likelihood of occurring			
Deposits	Liquidity adjustment deposits	Allow banks with surplus funds to deposit with BoK to limit excess volatility of call rate	1 business day	BoK policy rate - 50bp	N.A.

Source: BoK, Morgan Stanley Research

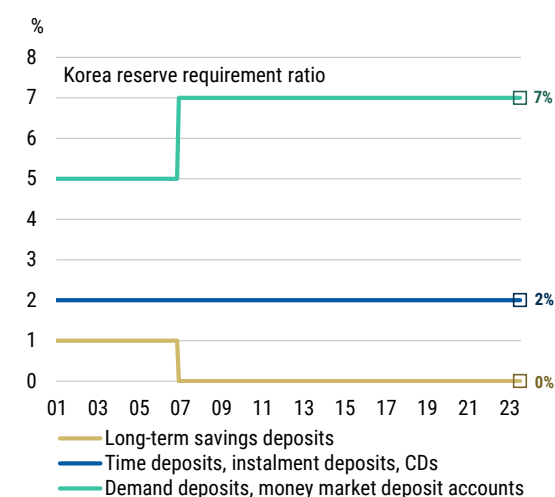
3. Reserve requirement ratio (RRR)

The reserve requirement ratio (RRR) is another tool which the BoK can use to influence the amount of liquidity in the financial system. Banks are required to hold a certain ratio of their liabilities as deposits with the BoK (or up to 35% in the form of vault cash). No interest is paid on these deposits.

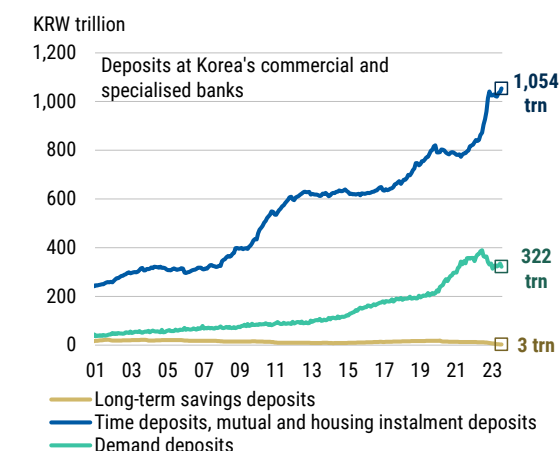
Higher RRR reduces liquidity: By raising the RRR, the BoK can increase the amount of reserves that banks have to keep at the central bank, thereby reducing the amount of funds available for lending and putting upward pressure on yields. Cutting the RRR would have the opposite effect of increasing liquidity and putting downward pressure on yields.

The RRR varies by deposit type: Since December 2006, the RRR for long-term savings deposits for housing and property formation have been set at 0%, while the RRR for time deposits, instalment savings, mutual instalments, housing instalments and CDs (excluding those issued to other commercial and specialised banks) have been set at 2%. The RRR for other deposit types, including demand deposits, has been set at 7% ([Exhibit 38](#)).

[Exhibit 39](#) gives an indication of the amount of deposits subject to the different RRR, though note there are some categories of deposits excluded due to data availability.

Exhibit 38: BoK's last RRR adjustment was in end-2006

Source: BoK, CEIC, Morgan Stanley Research

Exhibit 39: Most deposits are subject to the 2% reserve requirement ratio

Source: BoK, CEIC, Morgan Stanley Research

Bond Market Stabilisation Fund

A fund for exceptional circumstances: During periods of extreme market volatility, the authorities can also activate the Bond Market Stabilisation Fund. The fund was first created in 2008 during the GFC to invest in bank bonds, corporate bonds and commercial paper (CP), and was used again during the 2020 Covid pandemic and in late 2022 when credit market financing conditions tightened significantly.

Capital for the bond purchases comes from banks, securities firms, brokerages and insurance companies. In 2008, KRW 10 trillion of capital was injected into the fund. In 2020, KRW 3 trillion of capital was initially raised for the fund, of which KRW 1.4 trillion was used to buy corporate bonds and CP. In late 2022, the size of the Bond Market Stabilization Fund was set at KRW 20 trillion.

Other key bond products

KTB futures: With increased trading of KTBs and rising demand from investors to hedge against price fluctuations, the Korean government introduced KTB futures (3-year) in September 1999. There are currently three KTB futures markets by maturity: 3-year, 5-year and 10-year, with 3-year KTB futures being the most actively traded. The government is planning to introduce 30-year KTB futures.

KTB futures are traded on the exchange (KRX) and are cash settled (cash paid for the change in value at maturity). A standardised virtual bond is used as the underlying asset for KTB futures, in order to have consistent trading conditions of the underlying bonds such as time to maturity and coupon rate.

Interest rate swaps: Another tool that investors can use to hedge their interest rate risk is interest rate swaps (IRS). Korea has both a deliverable and a non-deliverable interest rate swap market (NIRS). In both markets, two parties agree to exchange a floating rate cash flow for a fixed rate cash flow (or vice versa).

The key difference is that the contract is settled in USD without any cash flows in KRW for the NDIRS. The 3m KRW certificate of deposit (CD) rate is the reference rate for the floating leg. Tenors for the interest rate swaps range from 1-year to 30-year.

How is the CD fixing set?

Waterfall method: Currently, the CD fixing is [set](#) by securities companies submitting asking prices on a voluntary basis. From October 2023, the CD fixing's calculation methodology will be changed to the waterfall method to increase its reliability.

Using qualified transactions: Under the waterfall method, the CD fixing is calculated by taking the average yield, excluding the highest and lowest values, of qualified CD transactions submitted by the top 10 securities companies with the largest CD transaction volumes in the past 12 months. Qualified transactions are CDs issued by AAA-rated commercial banks with 45-165 days' remaining maturity or 80-100 days' original maturity, and at least KRW 10 billion transaction volume.

Referring to adjacent issuance and distribution transactions: Should there be insufficient qualified transactions, the CD fixing will be calculated by referring to yields on adjacent issuance and distribution transactions, and taking the average yield, excluding the highest and lowest values.

Professional judgement: Alternatively, the CD rate will be calculated by applying professional judgement, such as referencing yields on similar bonds like the policy rate, yield on bank debentures or the trend in short-term money markets.

More details on the calculation methodology are available on the [KOFIA website](#).

Forward rate agreements (FRAs): A third instrument that investors can use to hedge their interest rate risk is forward rate agreements (FRAs), whereby two parties exchange fixed and floating interest payments for a specified period of time starting from a particular point of time in the future. Similar to IRS, the 3m KRW CD rate is the reference rate for the floating leg. FRAs are traded OTC.

KTB ETFs: KTB ETFs are funds that track KTB price moves. They can be traded on KRX KTB using investors' existing accounts for equities. KTB ETFs make it easier for retail investors to make bond investments as their minimum trade unit is usually smaller than the general trade unit of the KTB spot market. Their trade price is also almost similar to the market price for institutional investors, thus being favourable for retail investors who trade in small volumes.

Overview of MSBs

What are MSBs? MSBs are bonds that are issued by the BoK as part of its open market operations to manage liquidity in the financial system and steer the overnight interbank lending rate around the policy rate. They are commonly issued with maturities of 3 months, 1 year, 2 years and 3 years ([Exhibit 41](#)).

Towards the end of every month, the BoK will release the MSB auction schedule for the month ahead, which stipulates the auction dates and issuance amount for each maturity.

Exhibit 40: Amount of MSBs outstanding has fallen markedly since late 2020

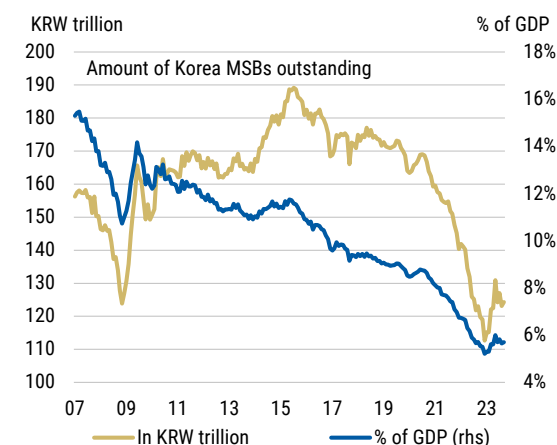
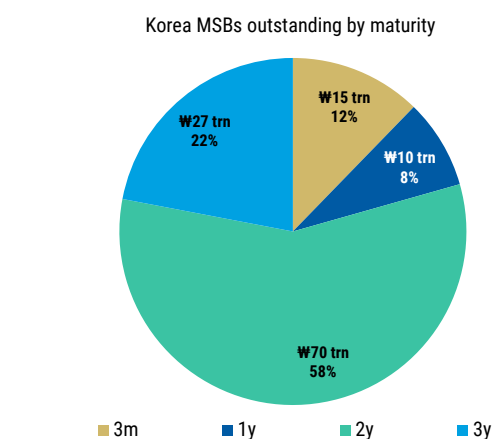


Exhibit 41: Most of the outstanding MSBs are of 2y maturity



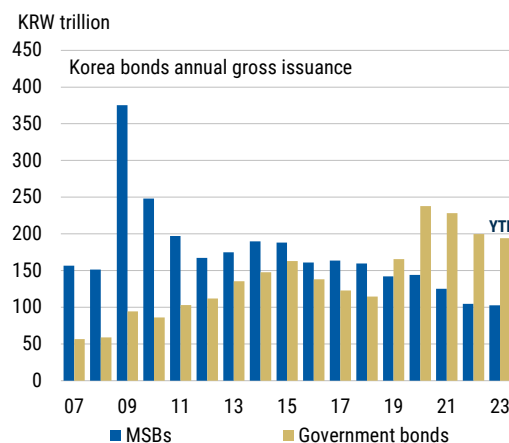
How big is the MSB market? The amount of MSBs outstanding currently stands at KRW 124 trillion (\$94 billion), or just 5% of the total size of Korea's domestic bond market ([Exhibit 40](#)). While this may look small, MSBs are in fact one of the most actively issued and traded bonds in the Korea market, next to government bonds.

Before the government pursued a more expansionary fiscal policy in 2019, the annual issuance of MSBs was even larger than that of government bonds, at ~KRW 160 trillion/year compared to ~KRW 115-140 trillion/year for government bonds ([Exhibit 42](#)).

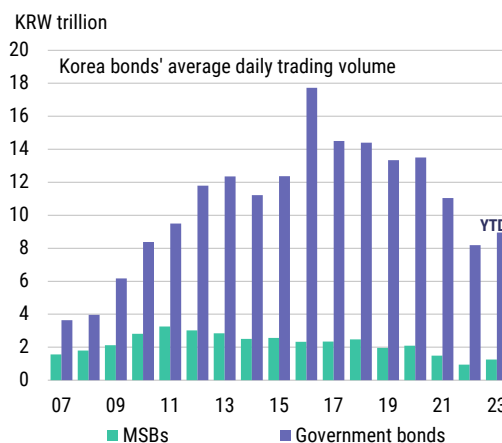
Why are MSBs issued so actively? It is unusual for a central bank's bond issuance to be so significant. One of the reasons for the large MSB issuance in Korea is that the government historically ran a fiscal surplus, hence reducing the need for government bond issuance.

A second reason for the meaningful MSB issuance is related to the fact that Korea has almost always run a current account surplus historically (see more details in [Trade flows](#)). Because of the natural inflows from the current account surplus, the BoK would have to issue MSBs to mop up the excess liquidity.

As the current account has narrowed significantly and volatility in the bond market has risen, MSB issuance has been reducing since 2021 ([Exhibit 42](#)).

Exhibit 42: MSB issuance was larger than that of government bonds before 2019

Source: KOFIA, Haver Analytics, Morgan Stanley Research

Exhibit 43: MSBs are among the most actively traded after government bonds

Source: KOFIA, Morgan Stanley Research

How liquid is the market? The average daily trading volume of MSBs is much smaller than that of government bonds, at KRW 1.3 trillion/day year-to-date, versus KRW 9 trillion/day for government bonds. However, compared to other domestic bonds, MSBs have one of the highest trading volumes ([Exhibit 22](#)).

Before 2021, MSBs were the most actively traded bonds, after government bonds, averaging ~KRW 2-3 trillion/day. As MSB issuance fell and bank bonds' issuance rose, the trading volume of MSBs fell below bank bonds, but there has been a renewed pickup in MSBs' trading volume this year. Year-to-date, the trading volume of MSBs has averaged ~KRW 1.3 trillion/day, comparable to that of bank bonds (KRW 1.4 trillion/day) ([Exhibit 43](#)).

How much do foreigners participate in this market? MSBs also have the highest foreign ownership next to government bonds. Foreigners own KRW 16 trillion of MSBs, or 13% of the market, just slightly less than the 21% foreign ownership in the government bond market ([Exhibit 44](#)). This is after foreign ownership of MSBs reduced markedly over 2022 with the global bond sell-off and lower issuance.

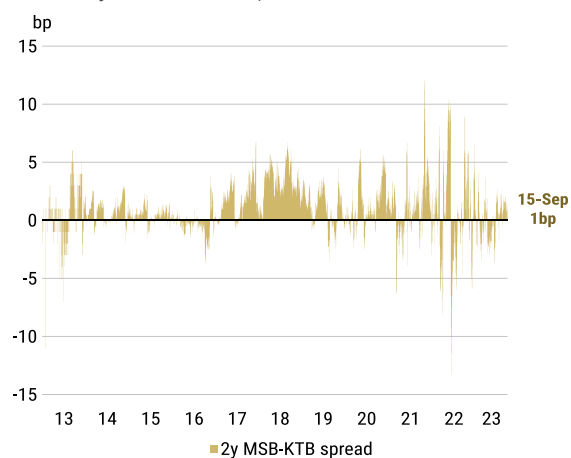
What is the yield on MSBs? MSB yields are comparable to those of KTBs, given that they are issued by the BoK. Under normal market circumstances, MSBs usually offer slightly higher yields than KTBs but not by more than 10bp ([Exhibit 45](#)).

Exhibit 44: Foreign ownership of MSBs is slightly lower than that of government bonds



Source: FSS, CEIC, Morgan Stanley Research

Exhibit 45: The yield advantage of MSBs over KTBS is usually less than 10bp



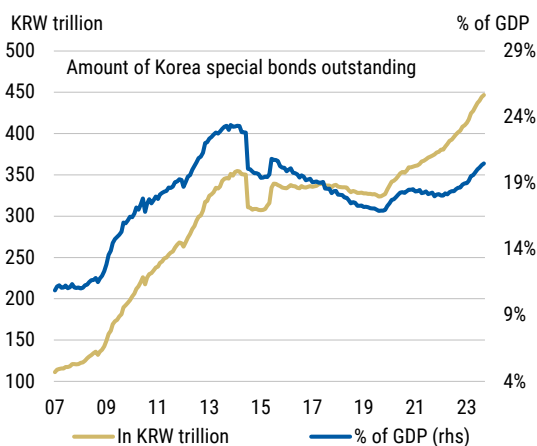
Source: Bloomberg, Morgan Stanley Research

Overview of special/policy bank bonds

What are special bonds? Special bonds refer to bonds that are issued by the nine state-owned corporations, namely Korea Electric Power Corporation (KEPCO), Korea Gas Corporation (KOGAS), Korea Expressway Corporation (KEC), Korea Land & Housing Corporation (KLH), Korea Railroad Corporation (Korail), Korea National Railway, Korea Water Resources Corporation (K-water), Korea SMEs & Startups Agency (KOSME), and Korea Deposit Insurance Corporation (KDIC).

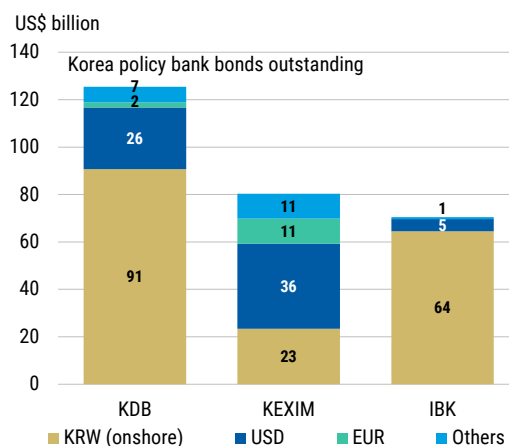
There is also a separate category of financial special bonds, which refers to bonds that are issued by the specialised banks such as the Korea Development Bank (KDB), the Export and Import Bank of Korea (KEXIM) and the Industrial Bank of Korea (IBK). Bonds issued by these three banks are also referred to as policy bank bonds and are usually classified under "bank bonds".

Exhibit 46: The special bond market has been growing since 2019



Source: KOFIA, Haver Analytics, Morgan Stanley Research

Exhibit 47: KDB and IBK have the largest amount of onshore bonds outstanding



Source: Bloomberg, Morgan Stanley Research

How big is the special bond... The size of the special bond market has been increasing since September 2019 and has reached KRW 447 trillion (~US\$337 billion), or 21% of GDP (Exhibit 46). This is up from the low of 17% of GDP in August 2019. After government bonds, special bonds are the next largest bond market in Korea, accounting for 16% of the total domestic bond market (Exhibit 20).

...and policy bank bond market? The amount of onshore policy bank bonds outstanding stands at KRW 225 trillion, of which KRW 113 trillion are issued by KDB, KRW 82 trillion are issued by IBK and KRW 30 trillion by KEXIM. There are also US\$98 billion (or KRW 119 trillion) of offshore policy bank bonds, mostly denominated in USD (Exhibit 47).

How liquid is the special bond... Special bonds are not traded as actively as government bonds, MSBs, bank bonds or other financial bonds. In recent years, the average daily trading volume of special bonds has been within the range of ~KRW 400-600 billion/day, with year-to-date volume picking up to KRW 686 billion/day (Exhibit 48).

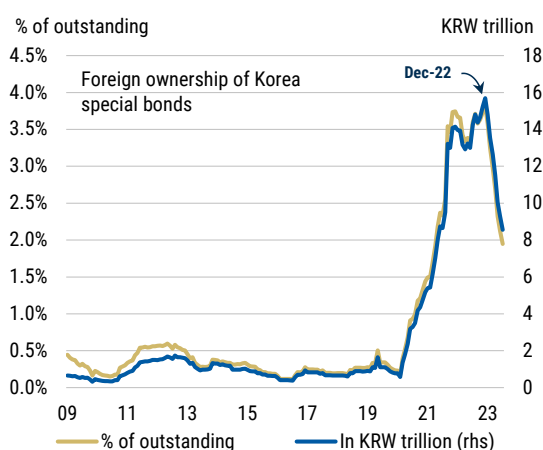
...and bank bond market? In comparison, the average daily trading volume of bank bonds (which includes other non-policy banks) is much higher, at KRW 1.4 trillion/day year-to-date. The average daily trading volume of bank bonds has been rising steadily from 2015-21 as the size of the market grew.

Exhibit 48: Special bonds are not traded as much as bank bonds



Source: KOFIA, Morgan Stanley Research

Exhibit 49: Foreign ownership of special bonds has been falling, after rising during Covid



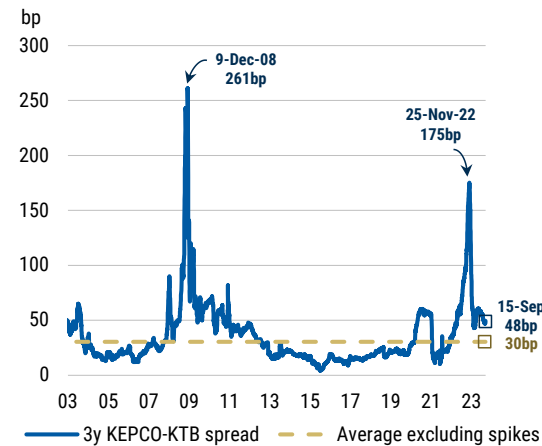
Source: CEIC, Haver Analytics, Morgan Stanley Research

How much do foreigners participate? Foreign ownership of special bonds is not as high as it is in government bonds or MSBs. Currently, foreigners hold ~KRW 8.6 trillion of special bonds, or ~2% of the market (Exhibit 49). This is down sharply from the peak of KRW 15.7 trillion, or 3.8% of total outstanding, at the end of 2022. Before the ramp up in foreign ownership of special bonds post-Covid, foreigners owned only ~0.2% of the market.

What is the yield advantage of special bonds over KTBs? Among special bonds, KEPCO has one of the largest amounts of bonds outstanding. Using KEPCO onshore bond yields as an indication, the 3-year KEPCO bond offers on average a ~30bp yield advantage over KTBs, under normal circumstances. Currently, its yield advantage is higher than the average, at 50bp, after tightening significantly from the 175bp seen during the height of the credit concerns last November (Exhibit 50).

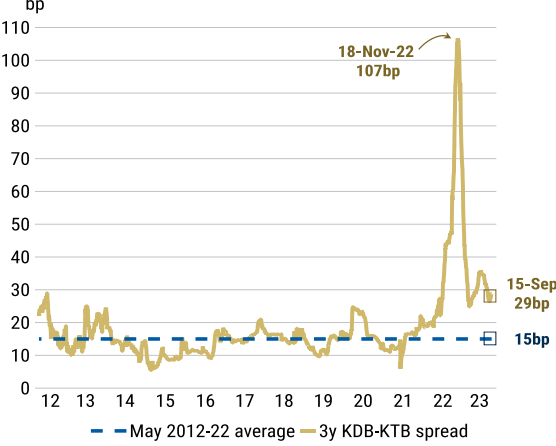
What is the yield advantage of policy bank bonds over KTBs? For policy bank bonds, KDB has the largest amount of bonds outstanding, so we use the 3-year KDB onshore bond yield as a guide. The 3-year KDB bond yields 15bp above KTBs on average, under normal conditions, less than the 3-year KEPCO bond. Currently, its spread over KTBs is at 29bp, again higher than the normal average. KDB spreads also widened materially during the credit market concerns in late 2022 ([Exhibit 51](#)).

Exhibit 50: KEPCO onshore bond spread over KTBs has tightened significantly



Source: KOFIA, CEIC, Morgan Stanley Research

Exhibit 51: 3-year KDB bond offers lower yield than KEPCO bond



Source: Bloomberg, Morgan Stanley Research

FX Market

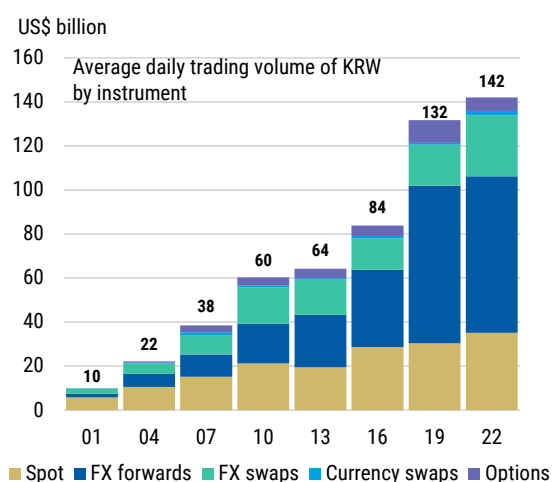
Gek Teng Khoo, Min Dai

Overview of the KRW FX market

Market structure: Korea has an onshore FX market and an offshore non-deliverable forward (NDF) market. The onshore market can be divided into the customer market and the interbank market. In the customer market, foreign exchange banks (local banks, foreign banks' branches in Korea and securities firms in Korea) deal with customers such as companies and individuals.

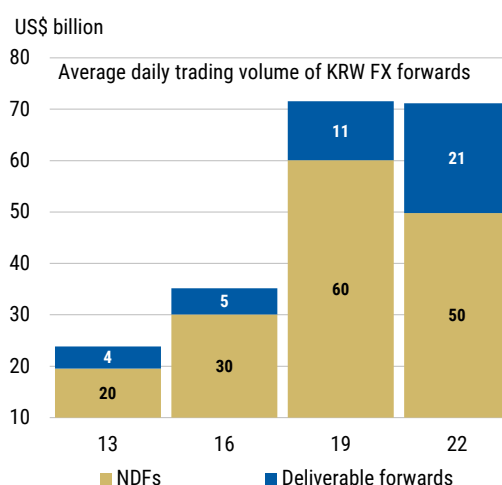
In the interbank market, foreign exchange banks deal among themselves. They may square the open positions arising from transactions in the customer market, so the two markets are interrelated. Most interbank transactions are traded through brokers, who intermediate between foreign exchange banks without holding any positions.

Exhibit 52: FX forwards account for half of KRW's total daily trading volume



Source: BIS, Haver Analytics, Morgan Stanley Research

Exhibit 53: NDFs make up 70-85% of total FX forwards trading volume



Source: BIS, Haver Analytics, Morgan Stanley Research

Most of the trading is done through NDFs: FX forwards, with an average trading volume of US\$71 billion per day, account for half of KRW's total daily trading volumes ([Exhibit 52](#)). Spot transactions account for a quarter of KRW's total daily trading volumes, with FX swaps constituting another 20%.

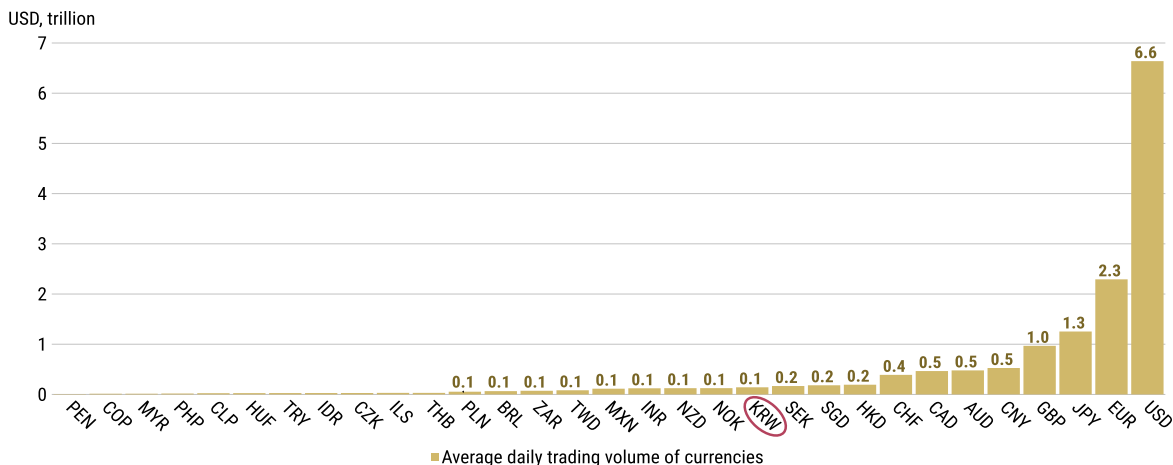
The dominance of FX forwards trading is due to limitations imposed on offshore investors in accessing the onshore KRW market. For example, offshore banks are not allowed to participate in the onshore interbank FX market, the onshore KRW spot market is only open for a limited number of hours, and documentation is required for cross-border fund transfers above US\$100,000 per year.

As such, non-deliverable forwards (NDFs) have become a popular tool for foreign investors to express their views on KRW or hedge their FX risk. Indeed, USD/KRW NDFs alone account for 70-85% of the total KRW forwards trading volume ([Exhibit 53](#)).

Overall, KRW is one of the most actively traded currencies across G10 and EM when looking at the combined trading volumes across all instruments ([Exhibit 54](#)). With an average daily trading volume of US\$142 billion, KRW is traded only less than CNY and the currencies of the financial centres (HKD and SGD) among EM currencies. It is even traded more actively than some G10 currencies like NOK and NZD.

Trading hours: The onshore FX spot market is usually open from 9:00am to 3:30pm KST on business days.

Exhibit 54: KRW is one of the most actively traded currencies



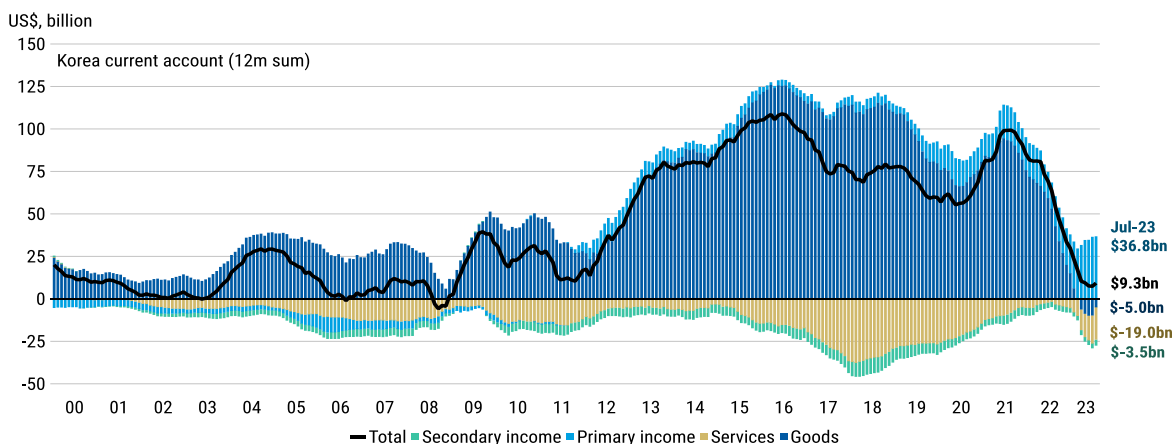
Source: BIS, Macrobond, Morgan Stanley Research

Trade flows

A source of USD supply: Being one of the largest exporters in the world, trade flows are an important source of demand and supply for KRW. Korea has almost always run a current account surplus since February 1998, mainly due to a large positive goods trade balance ([Exhibit 55](#)). This implies that trade flows are, on net, a source of USD supply for the economy.

A shrinking current account surplus: Korea's current account surplus has averaged ~3% of GDP since 2000. At the peak, in June 2016, the current account surplus stood at US \$109 billion (7.5% of GDP), with the services trade and secondary income deficits providing some offset to the US\$125 billion (8.6% of GDP) goods trade surplus ([Exhibit 55](#)).

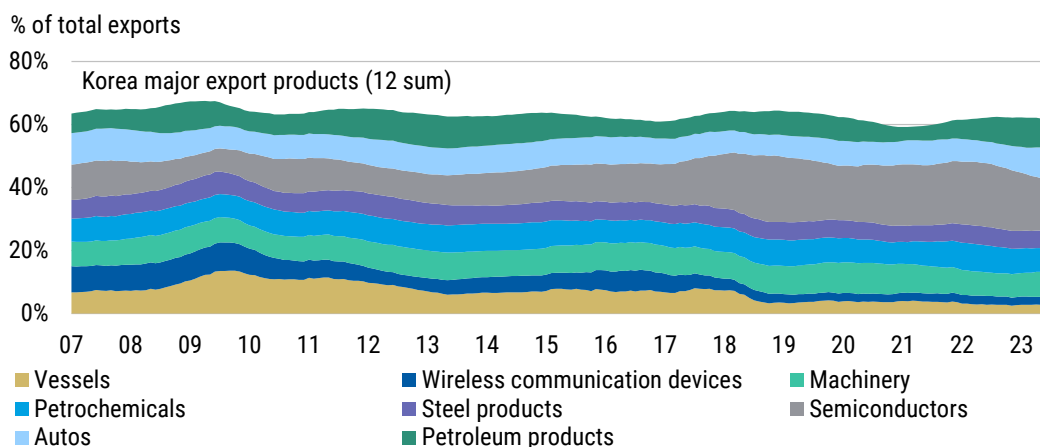
Since 2021, though, Korea's trade surplus has been shrinking rapidly and has turned into a ~US\$5 billion deficit. This was driven initially by imports rising faster than exports, before turning into a faster decline of exports last year as global growth slowed. This has brought the overall current account surplus down to just US\$9 billion (0.4% of GDP), the smallest since late 2008/early 2009.

Exhibit 55: Korea has almost always run a current account surplus

Korea's main exports: Despite the recent decline, Korea's goods exports (12-month sum) have increased from US\$172 billion (30% of GDP) in 2000 to \$626 billion (38% of GDP) today. Semiconductors are its biggest exports, accounting for 16-21% of total exports in recent years ([Exhibit 56](#)). Korea is also a large exporter of automobiles and auto parts (14% of total exports), petroleum products (8% of total exports), machinery (8% of total exports) and petrochemicals (7% of total exports).

Shipbuilders are a key USD supplier: A key export category that is often watched by investors in the FX market is the shipbuilders. While vessels' share of total exports has fallen over the years, it is still among Korea's top ten export products and Korea is one of the world's largest shipbuilders.

After securing orders, shipbuilders often sell USD/KRW forwards to hedge their FX risk, which can put downward pressure on the pair. In 1H23, Korea's shipbuilders [secured](#) US \$9.3 billion of orders, up 11.9%Y (1H22 orders: US\$8.2 billion).

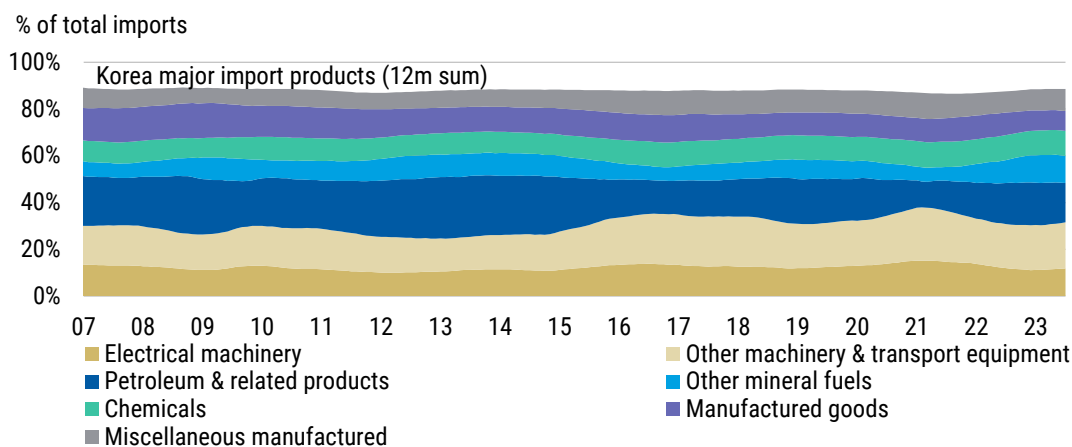
Exhibit 56: Semiconductors are Korea's main export product

Korea's main imports: On the imports side, Korea's goods imports have also increased from US\$160 billion (28% of GDP) in 2000 to US\$672 billion (41% of GDP) today. Petroleum and related products, as well as electrical machinery, are key import products, accounting for 17% and 12% of Korea's total imports respectively ([Exhibit 57](#)).

Energy importers are a key USD buyer: Korea is widely known to be one of the largest energy importers across the major EM and DM economies. A rise in energy prices would thus increase Korea's energy import bill, all else equal. Given energy products are often priced in USDs, this would result in more USD demand from the energy importers.

When oil prices were at their low in 2020, Korea's energy imports averaged US\$6.4 billion per month from March to December. This tripled to an average of US\$18.2 billion per month in 2022 following the rally in oil prices. With the decline in oil prices seen in 2H22, Korea's energy imports have been gradually declining to average US\$12.2 billion in the past three months (May-July 2023).

Exhibit 57: Petroleum & related products and electrical machinery are Korea's key import products

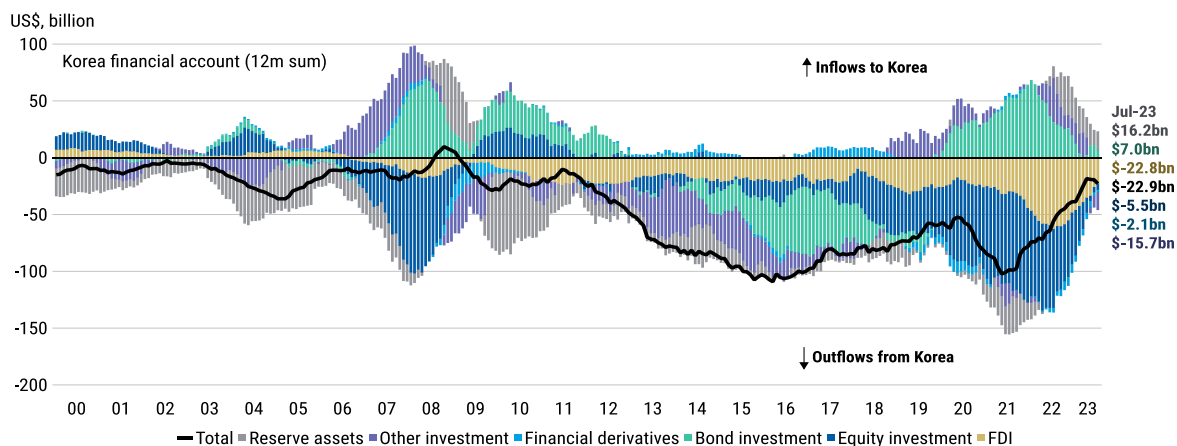


Source: Korea International Trade Association, CEIC, Morgan Stanley Research

Financial flows

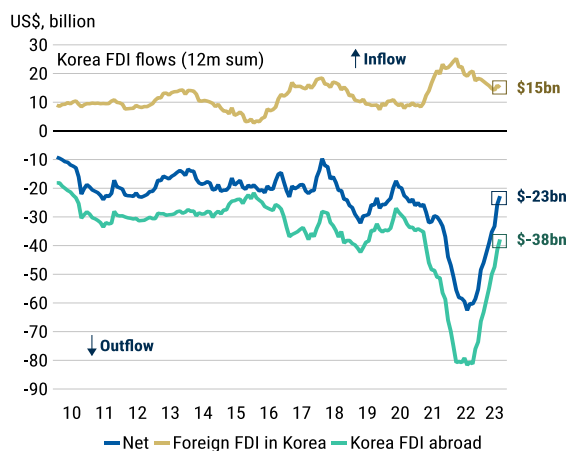
A source of KRW outflows: Given the persistent current account surplus, Korea has usually seen outflows in its financial account ([Exhibit 58](#)). Since 2000, Korea's financial account outflows have averaged ~3.2% of GDP. At the peak, the financial account deficit stood at 7.5% of GDP (US\$109 billion) in March 2016. But since 2H21, Korea's financial account outflows have been reducing to just US\$23 billion in July 2023, the smallest since late 2011, mainly due to lesser equity outflows.

FDI, portfolio and reserve asset flows are the key components: As seen in [Exhibit 58](#), the more persistent, larger sources of outflows in Korea's financial account are the foreign direct investment (FDI), equity and reserve asset flows. Long-term bond flows can also be sizeable but they oscillate between inflows and outflows.

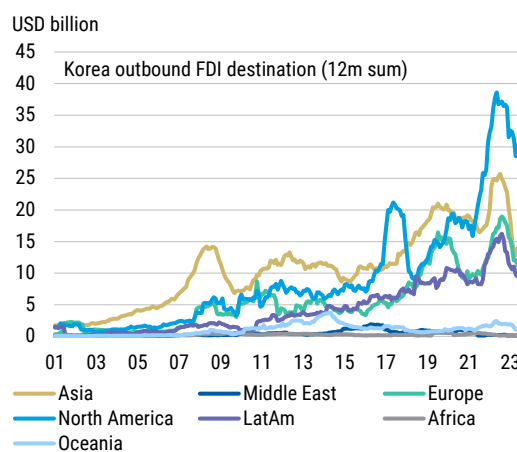
Exhibit 58: Korea has almost always seen financial account outflows

Source: BoK, Macrobond, Morgan Stanley Research

Key outflow source #1: Outbound FDI: Korea has seen net FDI outflows since 2006. The outflows were stable at ~US\$20 billion per year from 2010 to 2018 before increasing markedly after Covid to a peak of US\$63 billion in July 2022. Since then, the FDI outflows have reduced sharply to \$23 billion, converging towards levels seen in 2010-18.

Exhibit 59: Korea's net FDI flows are driven mainly by outbound FDI

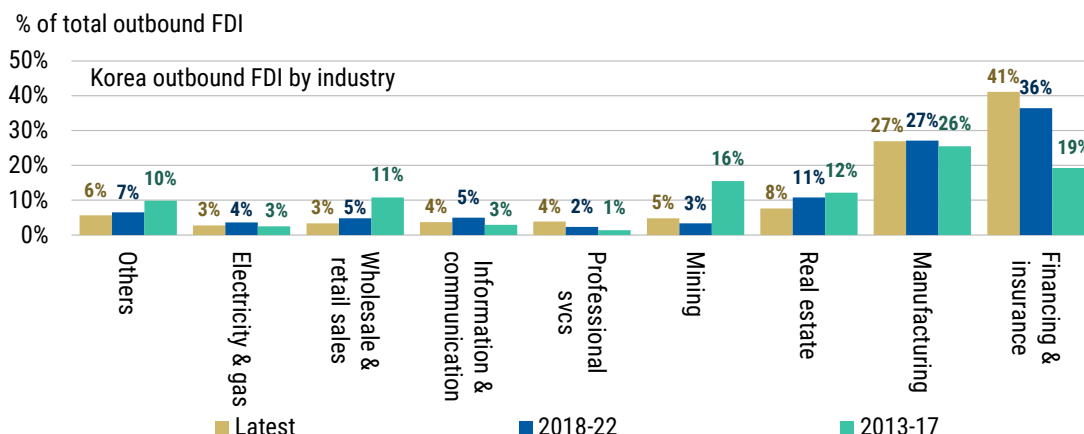
Source: BoK, Macrobond, Morgan Stanley Research

Exhibit 60: North America, particularly the US, is the main destination of Korea FDI

Source: KEXIM, CEIC, Morgan Stanley Research

The net FDI trend is determined mainly by Korea's investment in foreign companies, with foreign investment into Korean companies being comparatively smaller and more stable (Exhibit 59). Looking at the country breakdown, the US is by far the largest destination of Korea's outbound FDI, taking 42% of total outbound FDI. China and Vietnam each account for 4% of Korea's outbound FDI.

By industry, Korea's outbound FDI is concentrated in the financing & insurance and manufacturing sectors (Exhibit 61). These two sectors absorb 41% and 27% of Korea's total outbound FDI, respectively, and drove the swings in the FDI trend over 2020-22.

Exhibit 61: Financing & insurance and manufacturing are the main outbound FDI sectors

Source: KEXIM, CEIC, Morgan Stanley Research

Key outflow source #2: Foreign equity investment: On the portfolio flows side, Korea has seen net equity outflows almost all the time since 2013. Korea's net equity outflows have been increasing since September 2014 to peak at US\$88 billion in November 2021 – similar to the level seen during the February 2008 peak. Since then, equity outflows have eased significantly to US\$6 billion. Over the past two decades, *inflows* have peaked at ~US\$24-26 billion in both 2010 and 2004.

Prior to 2013, Korea's net equity flows were more correlated with foreigners' flows in Korean equities. But this changed, particularly from 2017, as Korean investors' foreign equity investments swelled ([Exhibit 62](#)).

NPS is a key domestic investor: The significant increase in Korea's foreign equity investments was partly due to the National Pension Service (NPS) [boosting](#) its overseas equity allocation. The NPS, with US\$746 billion of assets under management, is a key domestic investor (see [Overview of the Market](#) for more on the NPS's asset allocation).

Its flows can have an impact on FX as it invests mostly on an FX-unhedged basis. In 2015, the NPS [stopped](#) currency hedging on all but its bond investments, and started reducing its FX hedge ratio on its bond investments to reach zero by the end of 2018. Since then, the NPS has been investing abroad FX-unhedged, until December 2022 when the NPS [raised](#) its FX hedging limit to 10% for a limited period of time.

Exhibit 62: Korea's equity outflows have eased significantly since late 2021

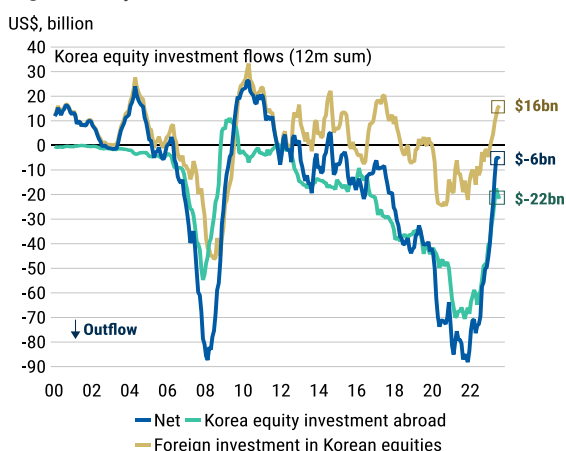
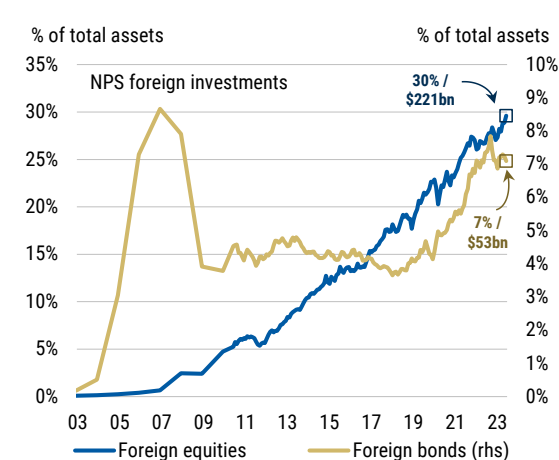


Exhibit 63: The NPS has been increasing its foreign investments



Key outflow source #3: FX reserves accumulation: Given Korea usually runs a trade surplus, it has been accumulating FX reserves, making reserve assets another persistent source of outflows in the financial account ([Exhibit 64](#)). Under normal economic circumstances, reserve asset outflows have usually been between US\$0-40 billion per year. But during extraordinary periods, e.g., during the 2008 GFC and the steep global central bank hiking cycle in 2022, it can turn into inflows.

Most of Korea's FX reserves are invested in government bonds (39.4%), with the rest allocated to agency bonds (14.1%), deposits (12.6%), ABS (11.5%), equities (11.4%) and corporate bonds (11%).

By currency, Korea's FX reserves are mostly invested in USD-denominated assets, with the USD allocation being higher than that of global FX reserves. While the share of USD in global FX reserves has been on a declining trend in recent years, Korea FX reserves' allocation to USD has been relatively stable around 67-70% and has even jumped higher to 72% in 2022 ([Exhibit 65](#)).

Exhibit 64: Reserve assets are usually a relatively stable source of outflows

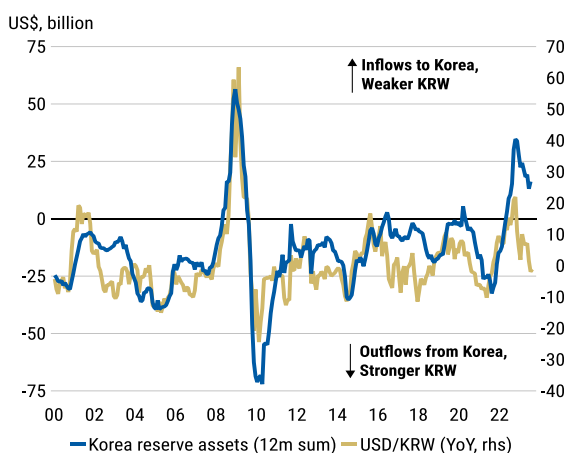
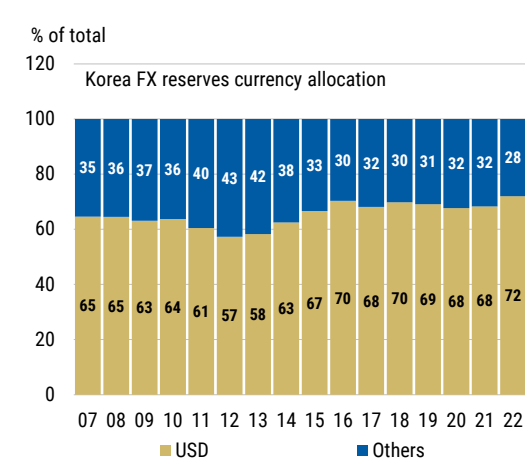


Exhibit 65: A large proportion of Korea's FX reserves are allocated to USD



How can the BoK/authorities smooth FX volatility?

Tailored measures: KRW is a free-floating currency so the exchange rate is determined by market demand and supply, but the authorities may intervene occasionally to smooth volatile moves. The market stabilisation measures may be tailored to the situation, e.g., in 2022 when the significant KRW weakness was driven by a widening trade deficit, the authorities **sought** ways to reduce energy imports as one of the measures.

Besides that, there are also a few main tools that they can use to smooth FX market volatility:

1. Verbal intervention: Commonly used commentary by the Finance Ministry and BoK officials include: the authorities are "**closely monitoring**" the FX market and are "**ready to take action to stabilise it if needed**", and the authorities are "closely monitoring" the market trend with "**extra alertness**".

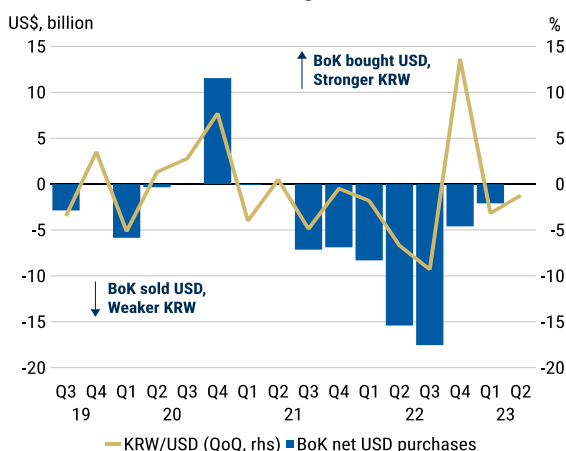
The officials could also state that they are "**preparing preemptive measures if needed**", or "**would take necessary measures in a timely manner**". During periods of particularly high volatility, the frequency of official comments could be higher.

2. FX interventions in the spot market: The BoK can also intervene directly in the markets. The BoK intervenes in both the spot and swap markets, with spot market interventions being used when the aim is to contain excessive market volatility and swap market interventions being used when the objective is to provide USD liquidity to the market.

The BoK's FX interventions are sterilised, i.e., it will issue or redeem MSBs to offset changes in the domestic money supply resulting from its FX interventions, keeping overall money supply stable.

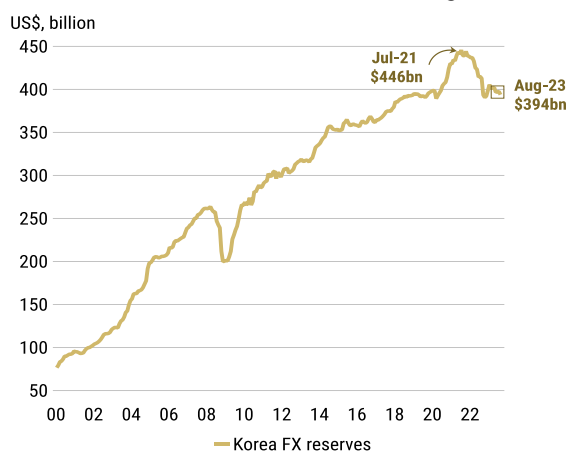
To increase transparency, the BoK started disclosing its net USD/KRW buying/selling from 2H18. As seen in [Exhibit 66](#), the BoK increased its USD selling last year as KRW weakened at a faster pace. The central bank's USD selling peaked at US\$17.5 billion in 3Q22 and has since eased to US\$2.1 billion in 1Q23.

Exhibit 66: BoK's USD selling has reduced



Source: Bloomberg, BoK, Macrobond, Morgan Stanley Research

Exhibit 67: Korea's FX reserves are still high



Source: BoK, Macrobond, Morgan Stanley Research

3. FX interventions via the swap market: The BoK can also provide USD liquidity to the

market via swaps to ease any tightness in USD liquidity. Historically, the BoK has established these swaps with the foreign exchange banks and NPS during extraordinary times. For example, the BoK supplied USD to banks via swap transactions in 2020 and 2008, tapping on its USD swap facility with the Fed.

Most recently, the BoK also opened a US\$35 billion FX swap line with the NPS in April 2023 to allow the state pension fund to obtain the USD it needs for its foreign investment from the BoK's FX reserves rather than the spot market. The BoK also had a similar FX swap line with the NPS in late 2022 (US\$10 billion) and in 2005-2008.

4. Foreign Exchange Stabilisation Fund: The government may also tap the Foreign Exchange Stabilisation Fund, which was set up in 1967 for the purpose of maintaining FX market stability. The FX Stabilisation Fund is included in the FX reserves managed by the BoK, but its funding and operations are decided by the MoEF.

In September 2022, the government used the FX Stabilisation Fund to absorb shipbuilders' USD forward contracts, so as to free up shipbuilders' credit limits with banks and allow them to sell more USD/KRW forwards for hedging.

5. FX macroprudential measures: The government can also adjust the FX macroprudential measures to influence foreign currency liquidity in the system. There are five key macroprudential measures:

- **Limits on the daily net FX positions that can be held by banks**
- **Limits on the daily FX derivatives positions that can be held by banks**
- **Foreign exchange stability levy** (levy that financial institutions have to pay on their amount of non-deposit foreign currency liabilities with less than one year's maturity)
- **Reserve requirement ratio for FX deposits** (foreign currency deposits that banks have to hold with the BoK as a proportion of their foreign currency liabilities)
- **Foreign currency liquidity coverage ratio** (high quality, liquid foreign assets that banks must hold to meet their expected net cash outflows over the next 30 days)

In March 2020, the government raised the FX futures trading limit, temporarily lifted the FX stability levy, and lowered the FX LCR to help release more USD to the market and ease banks' burden in getting foreign currencies.

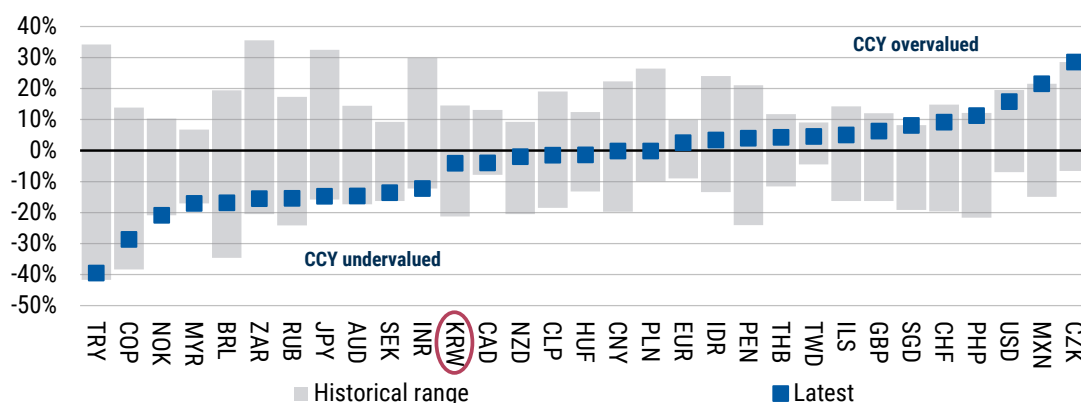
KRW valuations

Measuring long-term valuations using the BEER model: One common measure of FX valuations used is the Behavioural Equilibrium Exchange Rate (BEER) model, which suggests that the equilibrium real effective exchange rate (REER) is determined by a set of economic fundamentals that are carefully chosen based on a set of predetermined theoretical assumptions.

In the MS version of the BEER model, we stipulate that the fair value of a currency is determined by productivity growth differentials and terms of trade (see [Global Macro Strategy: Assessing Fair Value in FX: A BEER Approach](#), April 6, 2021, for more details).

Exhibit 68: KRW is most undervalued after MYR and INR in EM Asia

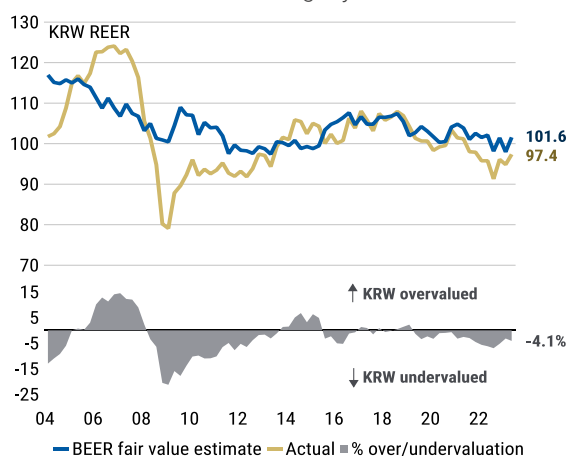
REER valuations based on BEER model



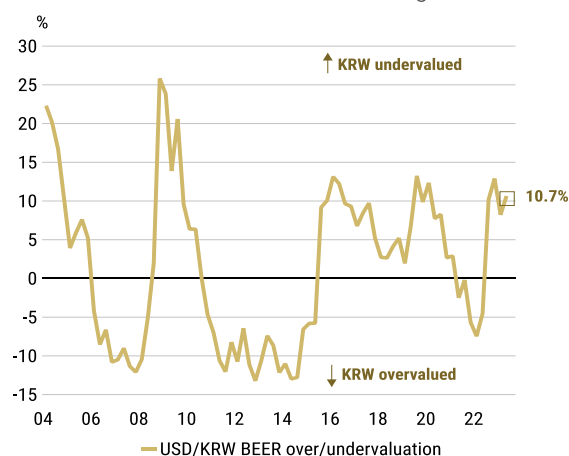
Source: Morgan Stanley Research

KRW is slightly undervalued: Based on this measure, KRW is slightly undervalued both on a REER basis and against USD, by 4% and 10% respectively (Exhibit 68). Post-GFC, KRW REER has been undervalued for most of the time, but the current level is slightly higher than the average 1% undervaluation over the past ten years. KRW REER's peak undervaluation most recently was at 7% in 3Q22.

Similarly, USD/KRW has been overvalued most of the time since 3Q15, with the magnitude of overvaluation averaging 3% over the past ten years. USD/KRW's overvaluation has come down from the peak of 13% in 4Q22 but remains higher than the 10-year average (Exhibit 70).

Exhibit 69: KRW REER is slightly undervalued

Source: BIS, Macrobond, Morgan Stanley Research

Exhibit 70: KRW is also undervalued against USD

Source: Bloomberg, Morgan Stanley Research

FX regulations and reforms

FX system governed by FETA: Korea's FX system and operations are governed by the Foreign Exchange Transactions Act (FETA). Under the Act, the MoEF is responsible for designing the overall FX policies. The BoK works with the MoEF to formulate the FX policies, and has the role of supervising the FX brokers and FX transactions, in addition to managing Korea's FX reserves.

Reporting of FX transactions: There are documentation or reporting requirements for certain types of FX transactions (more details available on [the BoK website](#)). For example, overseas remittances over the amount of US\$100,000 a year will [require](#) documentation, and firms borrowing more than US\$50 million of foreign currency in a year will need to report to the authorities.

Restricted access to the onshore market... Currently, as discussed in [Overview of the KRW FX market](#), foreigners' access to the onshore KRW market is also limited. Deliverable KRW is traded only on the onshore FX market, but firms based offshore that do not have branches in Korea are not allowed to participate directly in the onshore FX market.

...but reforms are on the way: As part of its efforts to get Korea included in global bond and DM equity indices, the government has proposed changes to improve foreigners' access to the onshore KRW market. These measures are expected to take effect from July 2024 at the earliest, after a six-month trial period. Some of the key measures [announced](#) are:

- **Opening up the onshore FX market:** Offshore financial institutions will be able to participate directly in the onshore KRW interbank market after obtaining registration from the Korean authorities. These Registered Foreign Institutions (RFI) will be able to trade deliverable FX spot, forwards and FX swaps, with both onshore and offshore customers. The RFI will trade through a local FX broker.
- **Extending the onshore FX market's trading hours** from the current 9:00am-3:30pm KST to 9:00am-2:00am KST, with plans to extend it to 24 hours eventually.
- **Developing market structure to be in line with global FX markets:** RFIs will be able to connect to the Application Programming Interface (API) that local FX brokers currently provide to local market participants. Aggregators, which are widely used FX trading platforms in the global market, will be introduced.

Exhibit 71: Opening up of Korea's onshore interbank FX market



Source: MoEF, Morgan Stanley Research

Equity Market

Joon Seok, Heewon Choi, Michelle Kim

Overview of the Market

Korea's secondary stock market can be divided into three parts: the KOSPI, KOSDAQ, and KONEX, which are all governed by KRX (Korea Exchange), a sole exchange operator.

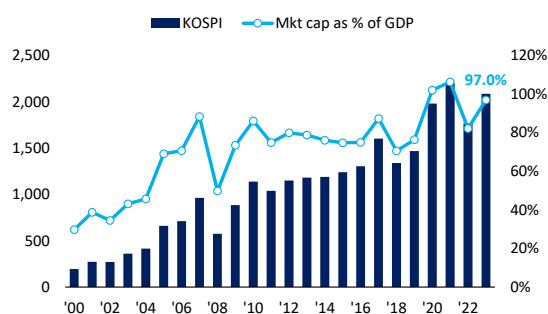
KOSPI (Korea Composite Stock Price Index): This is a benchmark index for Korean stocks, which is calculated based on a market cap-weighted method with a base value of 100 as of January 4, 1980. Total market capitalization of the KOSPI stood at W2,086tn (USD1.57tn) as of end-July, which is about 97% of nominal GDP. By sector, IT accounts for more than 40% of total capitalization, followed by consumer, energy & materials, and industrials.

- The KOSPI is relatively concentrated in large-cap stocks. The top 10 companies based on market cap accounted for 47.4% of the index as of July 31 – which is higher than 34.6% for the TOPIX and 42.7% for the TAIEX.

KOSDAQ (Korea Securities Dealers Automated Quotations): This is another major stock market more focused on smaller companies (SMEs) in the IT and Healthcare sectors. The KOSDAQ's market capitalization stood at W453tn (USD340bn) as of end-July, led by IT (43%) and Healthcare (21%).

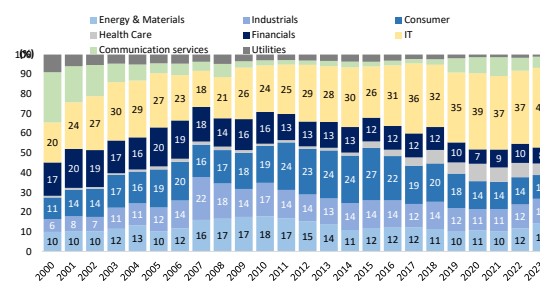
KONEX (Korea New Exchange): Similar to the KOSDAQ, this market is for early-stage SMEs but overall market cap is much smaller at around W4tn.

Exhibit 72: KOSPI: Market Cap vs. GDP



Note: Data as of July 31, 2023. Source: WiseFN, BoK, Morgan Stanley Research

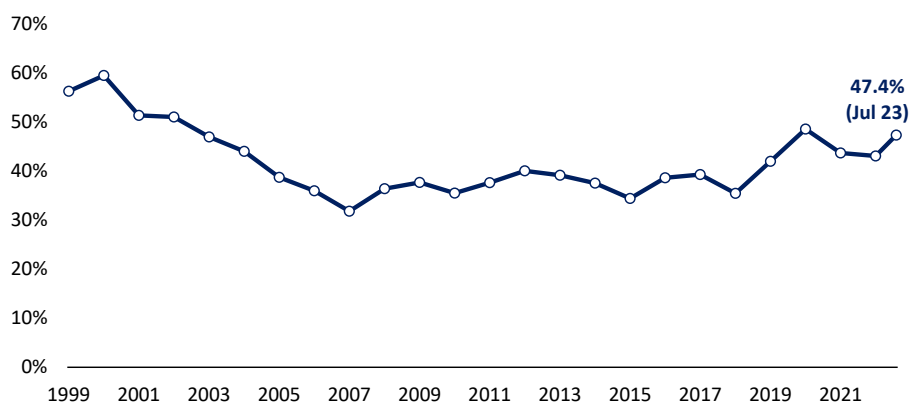
Exhibit 73: KOSPI: Market Cap by Industry



Note: Data as of July 31, 2023. Source: WiseFN, Morgan Stanley Research

Exhibit 74:

Proportion of Top 10 Large-cap Stocks within the KOSPI Based on Market Cap

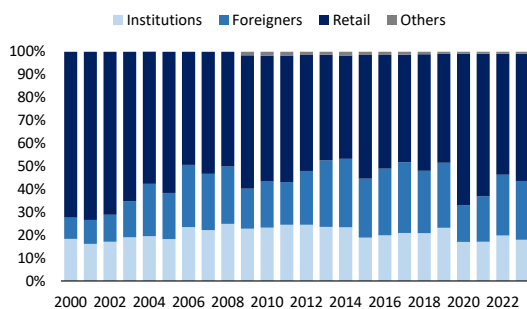


Source: WiseFN, Morgan Stanley Research

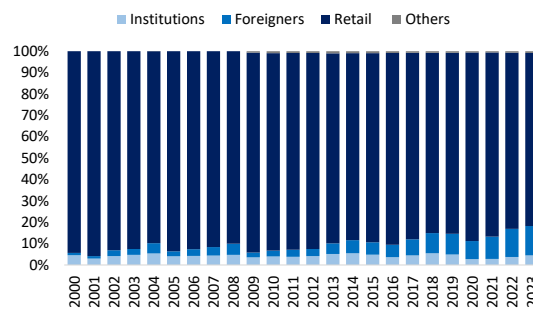
Market participation by type of investors

The KOSDAQ has higher retail participation than the KOSPI in terms of the proportion of retail investors within total trading value. Retail participation in KOSPI stocks significantly increased to the mid-60% range during the pandemic amid low interest rates, but it has fallen to the mid-50% range since 2022. Meanwhile, retail investors remain the dominant players in KOSDAQ stocks, with >80% of trading value coming from the retail side, though the number has recently decreased.

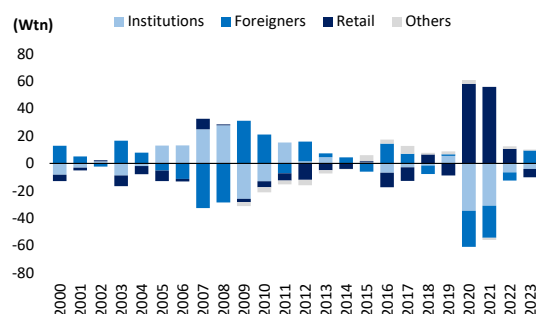
Foreigners and domestic institutions had been net sellers of KOSPI stocks for the past three years (2020-2022), but they have turned net buyers on a YTD basis as of end-July. On the other hand, foreigners and domestic institutions remained net sellers of KOSDAQ stocks while retail investors have been net buyers for the past 10 years since 2014.

Exhibit 75: KOSPI: Trading Value by Investor Type

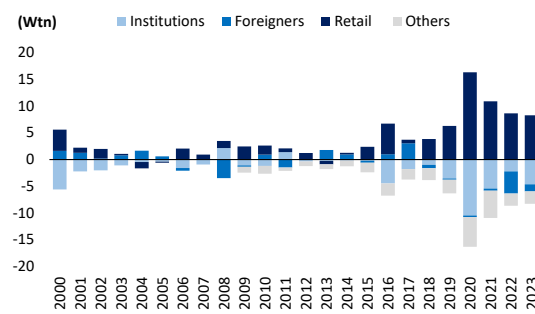
Note: Data as of Jul 31, 2023. Source: WiseFN, Morgan Stanley Research

Exhibit 76: KOSDAQ: Trading Value by Investor Type

Note: Data as of Jul 31, 2023. Source: WiseFN, Morgan Stanley Research

Exhibit 77: KOSPI: Net Purchase Value by Investor Type

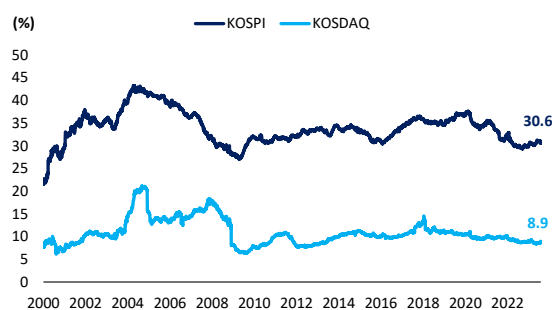
Note: Data as of Jul 31, 2023. Source: WiseFN, Morgan Stanley Research

Exhibit 78: KOSDAQ: Net Purchase Value by Investor Type

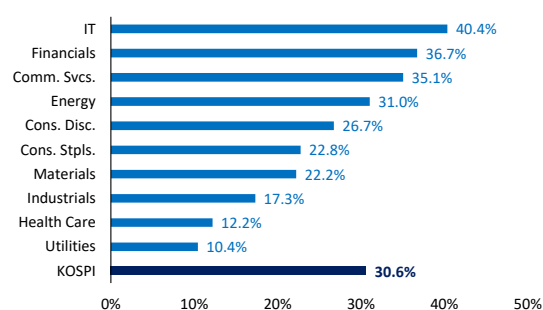
Note: Data as of Jul 31, 2023. Source: WiseFN, Morgan Stanley Research

(1) Foreign investors: Korea has opened its equity market to foreigners since January 1992. Foreign investors' share in the KOSPI (based on market cap) has moderated to the 30% range after it peaked at 42% in 2004. As of end-July, foreigners' shares in the KOSPI and KOSDAQ were 30.6% and 8.9%, respectively.

Despite the gradual opening of capital markets since 1992, still some industries are subject to foreign ownership limits. According to the rules set forth by the MoTIE (Ministry of Trade, Industry and Energy), industries such as communications services, broadcasting, newspapers, airlines, and power plants are subject to foreign ownership limits of up to 50%.

Exhibit 79: Foreigners' Share: KOSPI vs. KOSDAQ

Note: as of July 31, 2023. Source: WiseFN, Morgan Stanley Research

Exhibit 80: KOSPI: Foreign shares (%) by sector

Note: as of July 31, 2023. Source: WiseFN, Morgan Stanley Research

Exhibit 81:

KOSPI listed stocks subject to foreign ownership limits

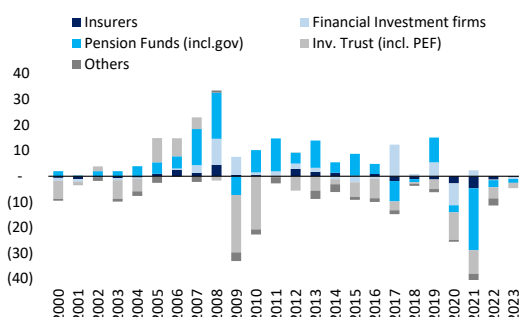
Sector	Company Name	Ticker	Market Cap (Wtn)	ADTV (Wbn, YTD basis)	Foreign Share Limit (%)	Foreign Share (%)
Airlines	Air Busan	298690.KS	0.3	3.5	49.99	0.4
	Asiana Airlines	020560.KS	0.8	3.1	49.99	6.3
	Jeju Air	089590.KS	1.0	6.3	49.99	4.3
	Jin Air	272450.KS	0.8	3.7	49.99	3.3
	Korean Air	003490.KS	9.1	27.4	49.99	16.2
	Korean Air (preferred)	003495.KS	0.0	0.1	49.99	1.2
Broadcasting /News	Tway Air	091810.KS	0.5	5.7	49.99	3.1
	Contentree JoongAng	036420.KS	0.3	3.3	30.0	1.3
	Hyundai Futurenet	126560.KS	0.3	1.1	49.0	1.5
	Hyundai Homeshopping	057050.KS	0.5	0.6	49.0	24.4
	KT Skylife	053210.KS	0.3	0.5	49.0	9.1
	LG HelloVision	037560.KS	0.3	1.4	49.0	6.4
Comm. Svcs.	SBS	034120.KS	0.5	4.7	-	-
	KT	030200.KS	7.7	33.1	49.0	39.5
	LG Uplus	032640.KS	4.4	11.0	49.0	38.9
Electricity Gas	SK Telecom	017670.KS	10.1	33.4	49.0	41.6
	KEPCO	015760.KS	12.3	26.7	40.0	14.6
	KOGAS	036460.KS	2.3	8.7	30.0	4.9

Note: Data as of July 31, 2023. Source: KOSCOM (Check+), Morgan Stanley Research

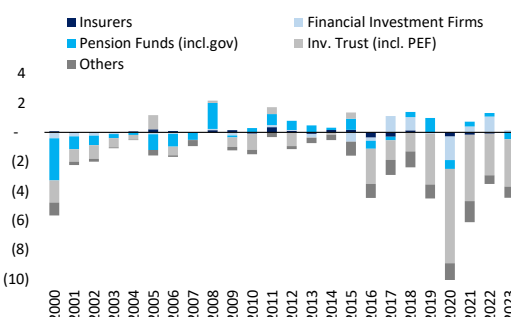
(2) Domestic institutions

According to KRX's investor classification scheme, domestic institutions can be divided into seven types:

- Financial investment firms: This type includes prop trading of brokers and asset managers.
- Insurers
- Investment trusts: Collective investment through public equity funds
- Private equity funds (including hedge funds)
- Pension funds: Funds owned by the government (including muni) are included in pension funds.
- Banks
- Other financial institutions

Exhibit 82: KOSPI: Net Purchase Value by Domestic Institutions

Note: as of July 31, 2023. Source: WiseFN, KRX, Morgan Stanley Research

Exhibit 83: KOSDAQ: Net Purchase Value by Domestic Institution

Note: as of July 31, 2023. Source: WiseFN, KRX, Morgan Stanley Research

NPS (National Pension Service): Among domestic institutions, the NPS plays an important role in the Korean market, with more than W140tn worth of AUM invested in the domestic equity market – as of end-May, the NPS held W145tn in domestic equities within total AUM of W973tn (14.9%). By sector, the NPS has highest exposures to IT given its 7.5% stake in SEC, which accounts for 20% of its total domestic equity portfolio,

followed by industrials and materials, etc.

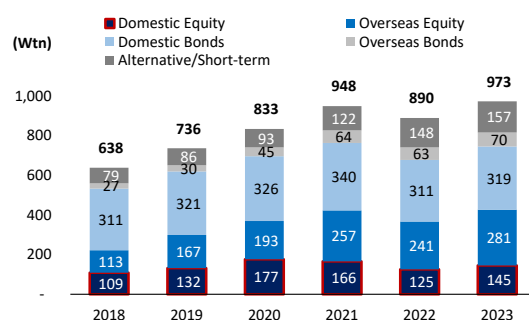
- In May every year, the NPS reviews its medium-term investment plan for the next five years. Once the medium-term targets are fixed, it revises its annual asset allocation plan. The NPS currently aims to increase the proportion of overseas equities and alternative assets from a medium-term perspective.

Exhibit 84: NPS: Investment Portfolio

(%)	2021	2022		2023		2024		2028
	Actual	Actual	Target	Actual*	Target	Scenario 1~2	Scenario 3	Target
Equity	44.6	43.8	44.1	43.8	46.2	48.4	48.3	55.0
Domestic Equity	17.5	14.9	16.3	14.9	15.9	15.4	15.4	
Overseas Equity	27.1	28.9	27.8	28.9	30.3	33.0	32.9	
Bonds	42.6	40.0	42.5	40.0	40.0	37.4	37.3	30.0
Domestic Bonds	35.9	32.8	34.5	32.8	32.0	29.4	29.3	
Overseas Bonds	6.7	7.2	8.0	7.2	8.0	8.0	8.0	
Alternative Assets	12.6	16.0	13.4	16.0	13.8	14.2	14.4	15.0
Private Investment			4.6		4.8	4.9	5.0	
Property			4.9		4.9	4.9	5.0	
Infrastructure			3.4		3.4	3.5	3.5	
Hedge Funds			0.5		0.7	0.9	0.9	
Total	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0

Source: NPS, Morgan Stanley Research

Exhibit 85: NPS: Total AUM



Note: as of May 31, 2023. Source: NPS, Morgan Stanley Research

Exhibit 86: NPS: Domestic Equity Portfolio

Breakdown by Sector

(%)	As of end-22
IT	30.3
Industrials	16.8
Material	10
Financials	9.4
Cons. Disc.	9.3
Healthcare	7.6
Comm. Svcs.	7.5
Cons. Stpls.	5.2
Energy, Utilities, etc.	3.9

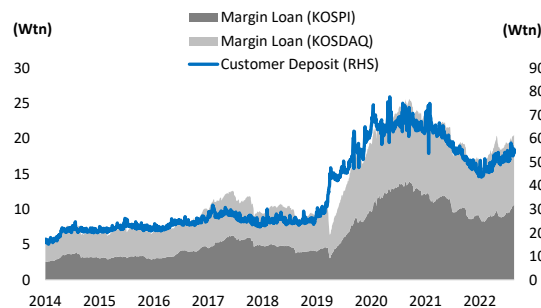
Source: NPS, Morgan Stanley Research

(3) Retail investors: Although the behavior of retail investors changes depending on market circumstances,, based on our view of the Korean equity market, we believe Korean retail investors tend to seek volatility and chase themes, with capital gains being the focus rather than dividend yields. Emotional considerations appear to play a bigger role in their decision-making compared to other investor types (foreign and institutional investors). Based on studies related to behavioral finance/investment, we see the following behavioral biases from retail investors:

- **Bandwagon effect:** This is closely linked to the fear of missing out (FOMO) and is usually most prevalent in times of high volatility. In that sense, Korea is well-connected via high accessibility of Internet and smartphones, and this makes individuals more prone to stories of high stock returns and market noise. This effect can also lead to herd behavior, effectively making them short-term momentum investors.
- **Confirmation bias:** Retail investors are in most cases disadvantaged with regards

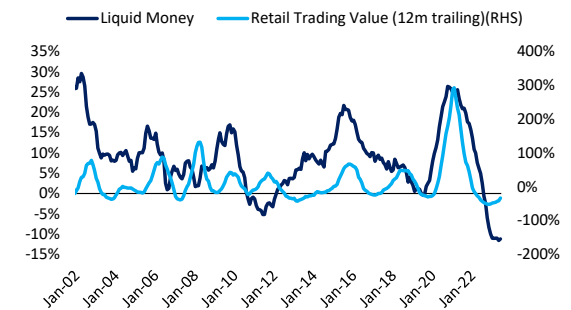
to information asymmetry, and they tend to search and favor information that confirms their existing beliefs but is not necessarily based on correct or solid reasoning. The bias becomes more widespread in volatile times as retail investors feel the need to rationalize sharp movements in their investments – and this could lead to an overconfidence bias if sharp positive returns are realized.

Exhibit 87: Customer Deposits vs. Margin Loan Balance



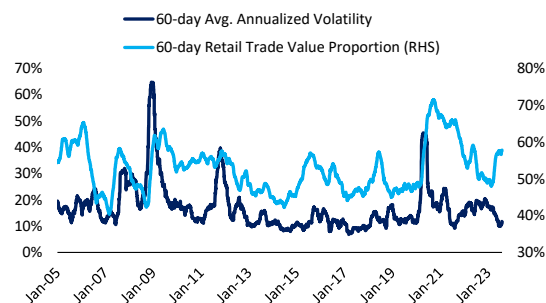
Source: KOFIA, WiseFN, Morgan Stanley Research

Exhibit 88: Liquid Money vs. Retail Trading Value



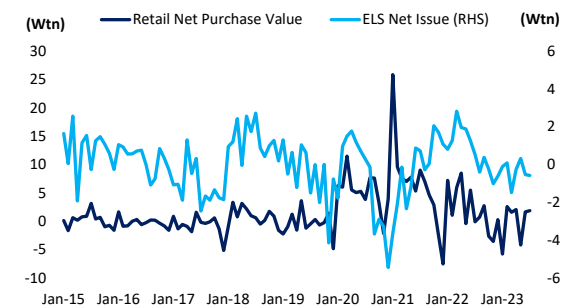
Source: BoK, WiseFN, Morgan Stanley Research

Exhibit 89: KOSPI: Annualized Volatility (60D) vs. Share of Retail Trading Value



Source: WiseFN, Morgan Stanley Research

Exhibit 90: Retail Net Purchase vs. ELS Issuance

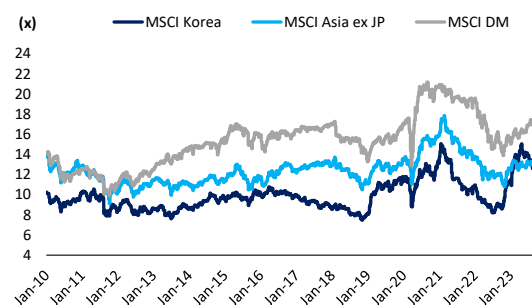


Source: KSD, Morgan Stanley Research

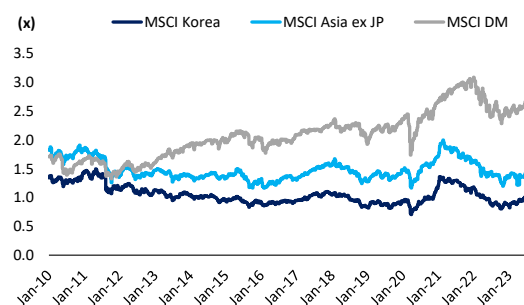
Valuation: Still Trading at a Discount to Peers

Korea continues to trade at a discount of around 20~30% to Axl on the basis of 12-month forward P/E and P/B. This discount to the rest of the region is often referred to as the "Korea Discount."

- Through 2013 to 2022, the 10-year average P/E on a 12-month forward basis for the KOSPI was 9.9x, 21% lower than that of Axl.
- Similarly, from a 12-month forward P/B standpoint, the KOSPI traded at a 31.1% discount on average to Axl during the same period.

Exhibit 91: MSCI Korea: 12-monh forward P/E

Source: MSCI, Datastream, Morgan Stanley Research

Exhibit 92: MSCI Korea: 12-monh forward P/B

Source: MSCI, Datastream, Morgan Stanley Research

Exhibit 93:

MSCI Korea vs. AxJ: 12-monh forward P/E discount

	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	Jul-23
MSCI Korea	-20%	-17%	-25%	-21%	-15%	-15%	-18%	-34%	-26%	-14%	-15%	-20%	-8%	1%
Energy	-5%	-28%	-14%	-5%	18%	-10%	-37%	-31%	-20%	-20%	61%	-9%	-35%	-4%
Materials	-16%	-13%	-16%	-13%	-16%	-16%	-21%	-21%	-12%	-17%	6%	-22%	-19%	9%
Industrials	-12%	-17%	-14%	-8%	-6%	12%	-4%	-9%	-3%	-7%	5%	-18%	-14%	19%
Cons. Disc.	-25%	-23%	-33%	-35%	-33%	-36%	-41%	-43%	-41%	-54%	-62%	-59%	-66%	-65%
Cons. Stpls	-16%	-15%	-15%	-17%	-10%	3%	-12%	-16%	-21%	-23%	-32%	-41%	-38%	-47%
Health Care	-23%	-2%	1%	-33%	-36%	40%	105%	130%	181%	94%	80%	8%	30%	29%
Financials	-24%	-25%	-25%	5%	0%	-5%	-4%	-19%	-27%	-33%	-45%	-41%	-45%	-49%
IT	-19%	-27%	-23%	-31%	-14%	-30%	-26%	-49%	-32%	-16%	-22%	-27%	26%	35%
Comm. Svcs.	-29%	-31%	-36%	-30%	-24%	-33%	-24%	-47%	-16%	11%	10%	14%	-1%	11%
Utilities	-19%	15%	50%	-28%	-67%	-41%	-57%	-50%	14%	2%	-20%	na	na	na

Source: MSCI, Datastream, Morgan Stanley Research

Exhibit 94: MSCI Korea vs. AxJ: 12-monh forward P/B Discount

	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	Jul-23
MSCI Korea	-22%	-18%	-24%	-24%	-29%	-31%	-27%	-33%	-34%	-37%	-30%	-28%	-36%	-30%
Energy	-20%	-32%	-24%	-28%	-39%	-1%	-17%	-13%	-22%	-28%	-11%	-12%	-36%	-27%
Materials	-17%	-18%	-24%	-25%	-35%	-28%	-34%	-33%	-36%	-46%	-27%	-41%	-41%	-17%
Industrials	-6%	-3%	-11%	-18%	-26%	-15%	-10%	-20%	-13%	-22%	-22%	-23%	-13%	-2%
Cons. Disc.	-30%	-25%	-34%	-38%	-40%	-44%	-52%	-59%	-65%	-74%	-76%	-62%	-70%	-66%
Cons. Stpls	-33%	-27%	-35%	-35%	-24%	-5%	-21%	-28%	-37%	-50%	-55%	-62%	-65%	-72%
Health Care	-40%	-9%	3%	-30%	-38%	20%	15%	95%	118%	59%	101%	37%	42%	38%
Financials	-31%	-34%	-44%	-34%	-39%	-40%	-42%	-42%	-47%	-55%	-60%	-53%	-54%	-58%
IT	-23%	-14%	-14%	-24%	-29%	-44%	-37%	-47%	-31%	-32%	-34%	-41%	-39%	-36%
Comm. Svcs.	-41%	-53%	-61%	-44%	-44%	-49%	-42%	-42%	-23%	-10%	-5%	0%	-32%	-46%
Utilities	-69%	-72%	-73%	-68%	-67%	-58%	-66%	-73%	-74%	-78%	-78%	-83%	-62%	-68%

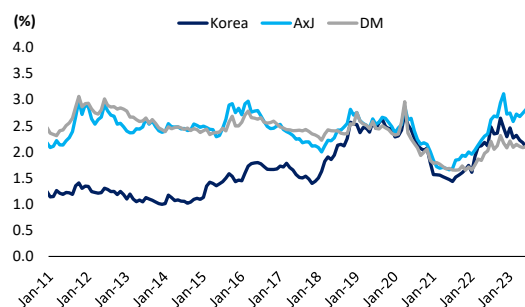
Source: MSCI, Datastream, Morgan Stanley Research

We believe that several factors contribute to the "Korea Discount," including: (1) lower shareholder returns, (2) lack of measures to protect minority shareholders and weaker corporate governance, (3) low data transparency and accessibility – not to mention the geopolitical risks from continued tension with neighboring countries.

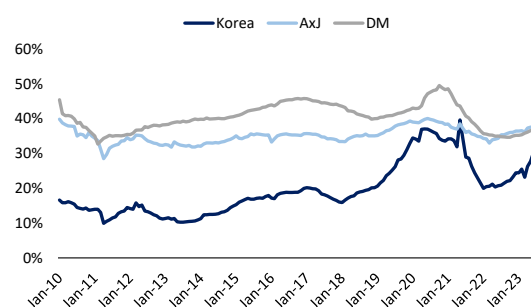
(1) Lower shareholder returns

Dividend payout is lower

The Korean market's dividend payout ratio of 24.4% in 2022 was well below that of other regions such as Asia ex-Japan (36.5%) and Developed Markets (35.2%). Korea's dividend yield of 2.1% as of end-July was lower than Japan's 2.24% and AxJ's 2.68%.

Exhibit 95: MSCI Korea vs. AxJ vs. DM: Dividend Yield

Source: MSCI, Datastream, Morgan Stanley Research

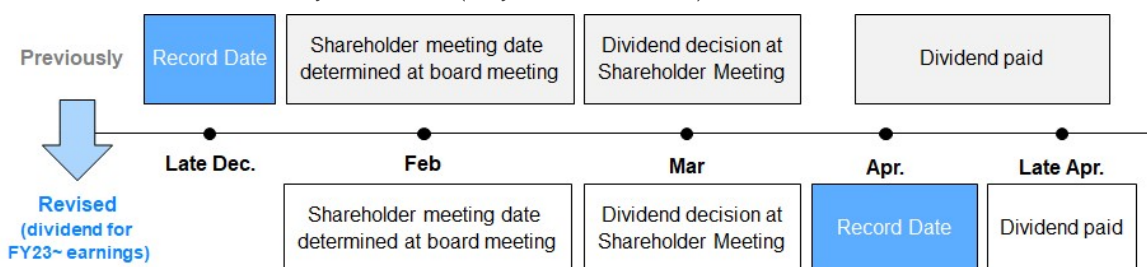
Exhibit 96: MSCI Korea vs. AxJ vs. DM: Dividend Payout Ratio (12-month trailing)

Source: MSCI, IBES, Datastream, Morgan Stanley Research

We expect an improving trend for dividend payout, given that companies will have more incentive to increase payout ratios in order to attract new investors following the recent changes in dividend payout process.

Dividend process reforms: Previously, Korean companies disclosed their year-end dividend amounts almost 2-3 months after the dividend record date (which is the last day of the fiscal year). The government has recently changed the rules for year-end dividends to bring Korean standards in line with international norms. Under the new rules, investors would now know the year-end DPS before the record date.

- Similar reforms for quarterly dividends are going to be made, as financial regulators plan to complete all the necessary revisions to the Capital Markets Act by end-2023.
- Major listed companies have already changed their articles of association so that the year-end dividend decision could precede the record date.

Exhibit 97: Korea: Dividend Payout Process (for year-end dividends)

Source: FSC, Morgan Stanley Research

Share buybacks are low and treasury share holdings are high

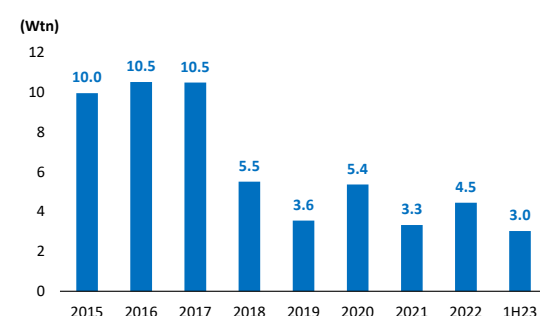
The current regulatory scheme does not limit the companies' treasury share holdings if the share buybacks are made within the annual distributable income. Share buybacks can be done in two ways:

1. Companies can buy their shares directly on the open market or through tender offers OTC (Over the Counter), or
2. They may entrust the amount allocated for share buybacks to a trustee who can buy the shares on behalf of the company.

Using a trust contract gives more flexibility compared to direct repurchases in terms of buyback period and amount. Direct repurchases should be done within three months after the BOD resolution, while the trust contract generally lasts for six to 12 months, and further extension of buyback period is allowed.

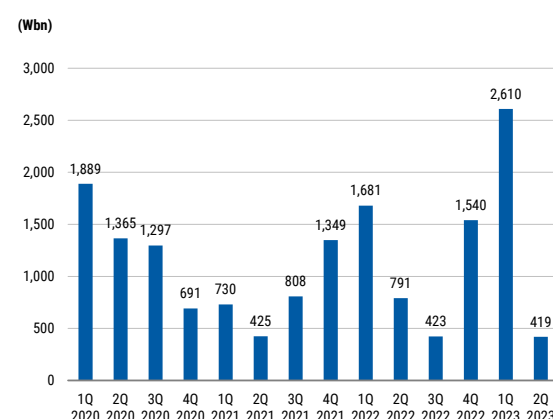
KOSPI-listed companies have increased their share buybacks, and we expect companies to be more active in share buybacks because we believe they will focus more on shareholder returns. However, there are still some cases where companies leverage their treasury shares to maintain or even increase control over new entities during spin-off processes. In order to regulate the unfair treatment of treasury shares, financial regulators have recently announced that additional measures will be released in 2H23.

Exhibit 98: KOSPI: Annual Share Buyback Trend



Note: Based on announcement date. Source: KRX, Morgan Stanley Research

Exhibit 99: Recent Share Buyback Announcement (KOSPI-Listed Companies)



Source: KRX, Morgan Stanley Research

Exhibit 100:

KOSPI and KOSDAQ: Number of companies holding treasury shares

Treasury Shares as % of Total Shares	KOSPI	%	KOSDAQ	%	KOSPI + KOSDAQ	%
0-5%	622	74.2%	1,382	85.3%	2,004	81.5%
5-10%	96	11.5%	161	9.9%	257	10.5%
10-15%	52	6.2%	45	2.8%	97	3.9%
15-20%	28	3.3%	17	1.0%	45	1.8%
20% and above	40	4.8%	16	1.0%	56	2.3%
Total	838	100.0%	1,621	100.0%	2,459	100.0%

Source: FSS, Morgan Stanley Research

(2) Minority shareholders not treated equally/lack of information transparency

Rights of minority shareholders can be protected by ensuring equal rights to participate in companies' decision making and equal access to information regarding company-related transactions. However, in Korea, minority shareholders have seen many cases where their rights were not appropriately protected, particularly in the case of M&A. In order to improve minority shareholder protection, the Korean government has recently developed the following measures:

- **Appraisal rights:** Minority shareholders were often excluded from the decision-making process when companies planned to restructure through split-offs, given

that there were no appraisal rights for dissenting shareholders. However, a new regulatory scheme on split-offs has come into effect since December 2022 with appraisal rights and tighter requirements on disclosures.

- **Mandatory tender offer in M&A process:** In the case of M&A where an acquirer becomes the largest shareholder by taking more than a 25% stake, the acquirer must make a cash tender offer for 50% + one shares held by minority shareholders.
- **Tighter disclosure requirements on insider trading:** The government has announced mandating a pre-disclosure in case a major shareholder (who owns a stake of more than 10%) or management plans to trade a stake of 1% or more (or worth W5bn or more) in the company. Current regulation only requires an ex-post disclosure, but companies should file pre-disclosures on the amount, pricing, and period of the transaction at least 30 days before the execution.

Aside from the policy reforms mentioned above, the government has announced additional initiatives to resolve structural issues that have been weighing on Korea's re-rating. These include mandating English disclosures, additional measures to protect minority shareholders, and FX market liberalization. Policy execution will be the key to successful reforms, which could mitigate the discount and lead to a re-rating of the Korean market.

Exhibit 101: Summary of Government-led Reforms to Tackle the "Korea Discount"

Key Initiatives	Measures	Details	Timeline
Improve Shareholder Return	Dividend Process Reform	Change dividend payout process for year-end dividend - Year-end dividend will be declared prior to record dates - Companies should revise their articles of association to adopt the changes	Feb 2023 ~
		Change dividend payout process for quarterly dividend - Quarterly dividend will be declared prior to record dates - Requires revisions to the Capital Markets Act	Pending in the National Assembly (Revisions proposed in Apr 2023)
		Include dividend process reform to key assessment measures for corporate governance report	2024 ~
Shareholder Protection	Tighten Rules on Disclosures	Tighter disclosures on share buybacks and disposal of treasury shares Tighten penalties on breaching disclosure rules for large shareholders	4Q23 4Q22 ~
	Mandatory tender offer	Mandate a tender offer in case an acquirer becomes the largest shareholder with more than 25% of stakes in an acquired company - An acquirer should buy at least 50% + 1 share of the acquired company - (Exceptions) If an acquisition is done for a special purpose (ex. industry restructuring), which will further be stated in relevant rules	Announced in Dec 2022 (FSC is preparing for revisions to the Capital Markets Act)
	Protect minor shareholders from split-off	Tighten disclosures on split-off process Appraisal rights for minority shareholders Tighten review process for IPO approval in case of split-off	4Q22 ~
	Incentivize Foreign Investment	Extend FX market business hours from current 9am-3.30pm to 9am to 2am (+1 day) Foreign Institutions eligible for RFI (Registered Foreign Institutions) can participate in FX spot / swap transactions Improve FX market infrastructure for settlement and payment scheme - Introduce a retail FX aggregator and allow non-residents to make FX transactions without bank accounts	2H24
	Mandate English Disclosures	Gradually increase the no. of companies required to report in English	2024 ~
	Abolish registration process	Replace the mandatory registration process with LEI and passport number	3Q24~

Source: FSC, MoEF, FSS, Morgan Stanley Research

Nuances of the Korean Equity Market

Short-selling bans

Short-selling was first introduced in 1969 with the advent of stock loans that enabled covered short-selling for retail investors, followed by stock lending for institutional investors in 1990s. Historically, temporary bans on short-selling were introduced in times of financial crisis.

The latest short-selling ban came into effect during the pandemic in 2020 (with exemptions for market-making activities). It was partially lifted on the constituents of the KOSPI 200 and KOSDAQ 150 in May 2021.

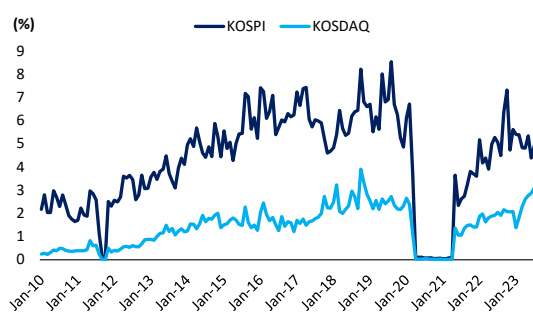
Exhibit 102: Changes in rules regarding short-selling

Feb-69	Launching of stock loans enabled covered shorts for retail investors
May-90	Covered short-selling was discontinued
Jan-94	Resumed covered short-selling for retail investors
Sep-96	Allowed covered short-selling for institutional investors
Apr-00	Banned naked short-selling due to the settlement failure
Oct-08	Temporary short-sell ban until May 09 due to the GFC
Aug-11	Temporary short-sell ban until Nov 11 due to the European sovereign debt crisis
Jun-16	Tightening of rules regarding disclosures on short-sell balance - Mandated the reporting in case the short-sell balance exceeds 0.01% of total issued shares
Mar-17	Introduced a temporary ban on overheated short-selling stocks - Stocks will be banned for short-selling on the next day if: - Share price drops by -3% or more on the day with short-selling value exceeding 30% of total trading value
Mar-20	Short-sell ban on the back of increasing market volatility with pandemic outbreak
Dec-20	Tightened punishments on unlawful short-selling (ex. naked shorts) - Fines (previously administrative penalties) will be levied on breaching of short-sell rules
May-21	Partially lifted the short-sell ban on constituents of KOSPI200 and KOSDAQ150
3Q22	- Tightened the designation rules of overheated short-selling stocks - Tightened disclosure requirements on short-sell positions lasting longer than 90 days

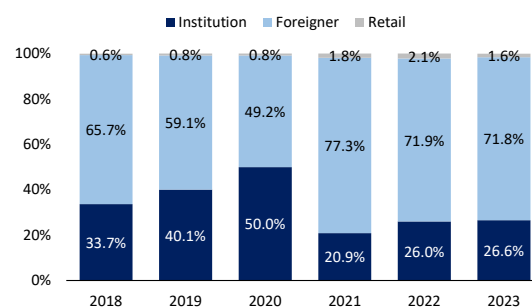
Source: FSC, KRX, Morgan Stanley Research

Recent regulatory updates regarding short-selling

- **Tighter monitoring and punishment on illegal short-selling:** Financial regulators have stiffened punishment for illegal short-selling with higher fines and imprisonment. The regulators have also announced that there will be fast-track investigations on naked short-selling.
- **Reforming rules on designation of overheated short-selling stocks:** Any stock whose share price has fallen more than 3% or for which the short-selling amount has more than doubled in a day will be designated as "overheated short-selling stocks" that are subject to a temporary short-selling ban for the next trading day, if the short-selling makes up more than 30% of total trading value.
- **Ensuring a level playing field for retail investors:** Financial regulators have lowered the margin required to short-sell stocks for retail investors to 120% from 140%, given the lower margin requirements at around 105~120% for institutional investors. Meanwhile, the maximum number days of opening short positions allowed for retail investors has also been extended to 90 from 60.

Exhibit 103: Short-selling value as a percentage of total trading value

Note: Based on monthly average. Source: WiseFN, KRX, Morgan Stanley Research

Exhibit 104: Short-selling by investor type

Note: Excluded data during the period subject to short-sell ban (from Mar 16, 2020 to Apr 30, 2021). Data as of Jul 31, 2023. Source: WiseFN, KRX, Morgan Stanley Research

Exhibit 105:

Designation of Overheated Short-Selling Stocks

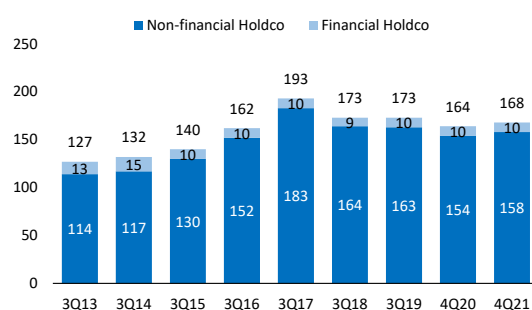
Market	Proportion of short-selling value	Negative Share Price Change	Increase in Short-sell Value (vs. average short-sell value for the past 40days)	Average proportion of short-sell value (for the past 40 trading days)
KOSPI & KOSDAQ	Higher than 30%	3% or above	2x or more	-
KOSPI	3x or more (vs. proportion of short-selling stocks for KOSPI-listed shares during the previous quarter)	5~10%	6x or more	-
	-	10% or above		-
KOSDAQ	3x or more (vs. proportion of short-selling stocks for KOSDAQ-listed shares during the previous quarter)	5~10%	5x or more	-
	-	10% or above		-
	-	-		5% or above

Source: KRX, FSS, Morgan Stanley Research

Holding companies

Korea's market has seen increasing numbers of companies adopting a holdco structure in response to tax incentives. The government has long pushed for holdco transitions as a part of its initiative to improve corporate governance by unwinding circular ownership within large corporate groups. In order to encourage holdco transitions, the government has provided the following tax incentives for holdcos:

- **Capital Gains Tax (CGT) deferral:** A company that has adopted a holdco structure through a share-for-share exchange can defer the capital gains tax until it sells the holding company's shares. The temporary CGT deferral will be allowed by the end of 2023.
- **Dividend exclusion rules:** Holding companies could get 80~100% tax exemption on dividend income received from their subsidiaries, based on previous dividend exclusion rules, which offered higher exclusion rates for holding companies vs. non-holdco. The government has changed the rule to provide the same exclusion rules for both holding and non-holding companies, which came into effect in January 2023.

Exhibit 106: Number of holding companies

Source: FTC, Morgan Stanley Research

Exhibit 107: Percentage of group affiliates owned by holding companies

Source: FTC, Morgan Stanley Research

Exhibit 108:

List of Holding Companies within Large Corporate Structures

Group	Holding Company		Subholding Company	
SK	SK	034730.KS	SK Innovation	096770.KS
			SK Sqaure	402340.KS
			SKC	011790.KS
	SK Discovery	006120.KS		
LG	LG	003550.KS		
	LX Holdings	383800.KS		
Lotte	Lotte Holdings	004990.KS		
GS	GS	078930.KS	GS Energy	Non-listed
HD Hyundai	HD Hyundai	267250.KS	KSOE	009540.KS
			HD Hyundai XiteSolution	Non-listed
NH	NH	Non-listed		
CJ	CJ	001040.KS		
Hanjin	Hanjin KAL	180640.KS		
Doosan	Doosan	000150.KS		
LS	LS	006260.KS		
DL	DL	000210.KS		
Booyoung	Booyoung	Non-listed		
Harim	Harim Holdings	003380.KQ	NS Shopping	Non-listed
HDC	HDC	012630.KS		
Hyosung	Hyosung	004800.KS		
Celltrion	Celltrion Holdings	Non-listed		
Taeyoung	TY holdings	363280.KS	Ecorbit	Non-listed
Kolon	Kolon	002020.KS		
SeAH	SeAH Holdings	058650.KS		
	SeAH Steel Holdings	003030.KS		
Hankook Tire	Hankook&Company	000240.KS		
Dongwon	Dongwon Industries	006040.KS		
HL	HL Holdings	060980.KS		
AmorePacific	AmoreG	002790.KS		
Samyang	Samyang Holdings	000070.KS		
Aekyung (AK)	AK Holdings	006840.KS		
Hite Jinro	HiteJinro Holdings	000140.KS		
Bando	Bando Holdings	Non-listed		
IS Dongseo	IS Holdings	Non-listed		
Nongsim	Nongsim Holdings	072710.KS		

Note: Data as of Dec 31, 2021. Source: FTC, Morgan Stanley Research

Preferred shares

A total of 114 preferred stocks are listed on the KOSPI with aggregate market capitalization of around W63tn, accounting for 3.2% of total market cap as of end-July. In Korea, preferred shares tend to trade at discounts to common shares. We cite the following factors:

1. Lack of voting rights,
2. Low liquidity, and
3. Low shareholder return with narrow gap between dividend yield on preferred vs. common shares.

In general, preferred shares are susceptible to dramatic share price volatility owing to lack of market depth with low liquidity.

In order to protect minority shareholders from any unfair trading activities that could artificially affect the price of preferred shares, financial regulators have tightened monitoring of price discrepancies between preferred and common shares. Moreover, they have tightened the delisting regulations on preferred shares, with higher hurdles for the market cap and the number of listed shares.

- Under the current regulatory scheme, preferred shares with less than 500k listed shares should be traded based on single-price auction scheme.
- Preferred shares with less than 200k listed shares or market cap of less than W200mn will be designated as "stocks with administrative issue," followed by delisting if they continue to fail to meet the requirements.

Exhibit 109: Regulations on Preferred Shares

KOSPI Market Listing Regulation (managed by KRX)	
Initial Listing	The number of stocks to be listed shall be at least 1mn
	Market cap shall be at least W5bn
	At least 25% of stakes should be held by the general shareholders
	The number of general shareholders should be at least 300
Designation of Stocks on Alert	The number of general shareholders becomes less than 100 as of FY-end
	The number of listed shares is less than 200,000 as of the end of semiannual period
	Monthly average trading volume in the semiannual period is less than 10,000
	Market cap being less than W2bn for 30 consecutive trading days
Delisting	In case the common shares of the issuer are delisted
	The number of general shareholders continues to be less than 100 as of FY-end while designated as stocks on alert.
	The number of listed shares continues to be less than 200,000 as of the end of the semiannual period while designated as stocks on alert
	Monthly average trading volume in the semiannual period continues to be less than 10,000 while designated as stocks on alert
	Market cap for 90 days after being designated as stocks on alert fails to meet any of below requirements
	(a) Market cap should be at least W2bn for at least 10 consecutive trading days (b) The number of days with Market cap being at least W2bn should be at least 30 days

Source: KRX, Morgan Stanley Research

Exhibit 110: Listing and Delisting Regulations on Preferred Shares (last updated in Jul 2020)

Requirements	Initial Listing		Delisting	
	Previous	New	Previous	New
Number of listed shares	At least 500 thou. shares	At least 1mn shares	Less than 50 thou. shares	Less than 200 thou. shares
Market Cap	At least W2bn	At least W5bn	Less than W500mn	Less than W2bn

Source: FSC, KRX, Morgan Stanley Research

KOSDAQ to KOSPI migration

There have been some cases where KOSDAQ-listed companies migrate to the KOSPI. there is no specific rule that regulates a KOSDAQ-listed company that seeks to migrate to the KOSPI, but the stock should meet the initial KOSPI listing requirements.

Listing requirements: KOSPI vs. KOSDAQ

The KOSPI and KOSDAQ have both quantitative and qualitative requirements for initial listing.

- Qualitative criteria evaluate management's continuity and capability, as well as the corporate governance and shareholder protection.
- Meanwhile, quantitative criteria stipulate the requirements of financial performance and minority shareholders.

The KOSPI has tighter requirements for both financial performance and minimum distribution compared to KOSDAQ. A minimum operating history of three years is required for the KOSPI.

Exhibit 111: KOSPI: Quantitative Requirements for Initial Listing

Operating History		At least 3 years
Size (Both A and B)		A. No. of common shares to be listed should be 1mn or above B. Capital should be at least W30bn
Financial Performance (Either one of the five criteria)	I. Either 1 or 2	1. Revenue of at least W100bn for the latest fiscal year, and average revenue of at least W70bn for the past 3 years 2. Either A, B or C A. Pre-tax profit should be at least W3bn for the latest fiscal year and aggregate of W6bn for the past 3 years B. ROE should be at least 5% for the latest fiscal year and 10% for the past 3 years
		Revenue should be at least W100bn for the latest fiscal year with market cap at the timing of application for IPO should be at least W200bn
	II.	Pre-tax profit for the latest fiscal year should be at least W5bn with market cap at the timing of application for IPO should be at least W200bn
	III.	Market cap at the timing of application for IPO should be at least W500bn, with capital of at least W150bn
	IV.	Market cap at the timing of application for IPO should be at least W1tn
Minimum Distribution	I.	A. Publicly held shares should be at least 5mn or 25% of total shares B. Publicly offered shares should be at least 5mn or 25% of total shares C. Shares that are publicly offered after the submission of the application for eligibility review should be at least 10% and meet either of the following criteria: i. 1mn shares for the companies with capital of W50bn to W100bn or with market cap of W100bn to W200bn ii. 2mn shares for the companies with capital of W100bn to W250bn or with market cap of W200bn or W500bn iii. 5mn shares for the companies with more than W250bn of capital or W500bn of market cap
		General shareholders should be at least 500

Source: KRX, Morgan Stanley Research

Exhibit 112: KOSDAQ: Quantitative Requirements on Initial Listing

Financial Performance (Either I or II)	I. Any one of the following 1 through 4	1. Pre-tax profit for the latest fiscal year should be at least W5bn 2. Pre-tax profit for the latest fiscal year should be at least W2bn, with minimum market cap of W9bn 3. Pre-tax profit for the latest fiscal year should be at least W2bn, with minimum capital of W3bn 4. Revenue for the latest fiscal year should be at least W10bn with positive pre-tax profit and minimum market cap of W20bn
		1. Market cap of W100bn 2. Market cap of W50bn with market cap to equity book value ratio of 200% or higher 3. Market cap of W50bn and revenue of W3bn for the latest fiscal year with average revenue growth at 20% for the past two years 4. Market cap of W30bn and revenue of W10bn for the latest fiscal year
	II. Any one of the following 1 through 6	5. Minimum capital of W25bn 6. (For KOREX-listed companies) Market cap of W75bn with ADTV of W100mn for the past 12months, and at least 20% of shares should be held by minority shareholders
		I. Minority shareholders should be at least 500, and meet any one of the following criteria: a. Shares that are publicly offered after the submission of the application for eligibility review should be at least 10% in case the minority stakes are less than 25% b. Shares that are publicly offered after the submission of the application for eligibility review should be at least 5% in case the minority stakes are 25% or higher
	II. Minority stakes are at least 25%, and the shares publicly offered after the submission of the application for eligibility review should meet both A and B: A. At least 10% of total shares B. Any one of the following 'a' through 'c' a. 1mn shares for the companies with capital of W50bn to W100bn or with market cap of W100bn to W200bn b. 2mn shares for the companies with capital of W100bn to W250bn or with market cap of W200bn to W500bn c. 5mn shares for the companies with more than W250bn of capital or with market cap of W500bn	

Source: KRX, Morgan Stanley Research

Implications for share price performance: No major trends have been seen from the past examples of migration from KOSDAQ to KOSPI, but in general, large-cap stocks appeared to perform better around the migration period.

Exhibit 113: KOSDAQ to KOSPI: Share price performance around the migration

Company Name	Ticker	Date of Migration to KOSPI	Abs. Perf					Rel. Perf				
			-3 month	-1 month	+1 month	+3 month	+12 month	-3 month	-1 month	+1 month	+3 month	+12 month
Coway	021240.KS	8/7/2001	-21.6%	-3.7%	-7.4%	-0.2%	251.4%	-23.6%	1.8%	4.8%	0.9%	231.6%
NC Soft	036570.KS	5/22/2003	-22.4%	-20.5%	1.5%	54.1%	162.2%	-23.8%	-21.9%	-13.8%	30.3%	130.2%
Kangwon Land	035250.KS	9/4/2003	-3.8%	2.0%	5.3%	13.7%	0.7%	12.1%	7.5%	11.7%	8.0%	-6.7%
IBK	024110.KS	12/24/2003	-3.4%	10.1%	10.4%	19.4%	9.6%	6.5%	15.1%	1.8%	10.1%	-1.1%
Asiana Airlines	020560.KS	3/28/2006	29.9%	10.3%	-1.5%	-12.8%	-45.0%	18.5%	9.2%	-8.7%	-13.7%	-17.7%
LG Uplus	032640.KS	4/21/2006	-10.5%	-12.4%	11.5%	-11.9%	3.0%	0.2%	-3.8%	8.9%	4.3%	28.8%
Bookook Steel	026940.KS	10/14/2006	1.4%	3.9%	-34.7%	2.7%	0.9%	-8.8%	-4.2%	-14.3%	18.2%	-19.7%
NAVER	035420.KS	11/28/2006	22.5%	-7.8%	-1.6%	13.1%	48.2%	-14.6%	2.2%	-5.5%	15.0%	6.5%
Kiwom Securities	039490.KS	8/3/2009	-1.1%	-15.2%	-12.3%	-20.7%	-17.9%	9.6%	-5.9%	-15.4%	-21.7%	-32.3%
Shinsegae Food	031440.KS	4/29/2010	-17.4%	-1.6%	-11.2%	-1.4%	8.6%	-10.1%	0.0%	-5.1%	-4.0%	-18.2%
Muhak	033920.KS	7/20/2010	-8.6%	-15.0%	-2.1%	18.8%	144.3%	-9.3%	-13.6%	-4.6%	10.8%	120.2%
Able C&C	078520.KS	9/7/2011	12.3%	-18.0%	-10.7%	-4.6%	217.0%	-0.7%	-20.0%	-6.7%	-8.3%	214.4%
Hana Tour	039130.KS	11/1/2011	45.6%	-0.5%	-3.3%	1.2%	62.0%	37.4%	6.8%	-3.6%	-0.4%	61.9%
KOREIT	034830.KS	7/11/2016	-10.6%	-1.2%	-11.8%	-11.8%	-8.3%	-10.2%	-2.7%	-14.6%	-15.0%	-28.8%
Dongseo	026960.KS	7/15/2016	-2.2%	-1.8%	-4.7%	-14.1%	-14.4%	-2.1%	0.6%	-6.4%	-14.0%	-34.1%
Kakao	035720.KS	7/10/2017	-16.0%	7.6%	6.4%	41.2%	14.7%	-5.1%	7.7%	6.9%	40.7%	18.4%
Celltrion	068270.KS	2/9/2018	-40.0%	2.8%	13.2%	-10.1%	-24.0%	-47.6%	-3.0%	9.1%	-14.3%	-16.1%
DoubleU Games	192080.KS	3/12/2019	-8.7%	-7.3%	0.0%	10.3%	-27.4%	-5.3%	-8.2%	-3.1%	13.0%	-15.9%
POSCO FutureM	003670.KS	5/28/2019	27.2%	4.5%	1.7%	-7.8%	9.1%	18.1%	-1.8%	-2.5%	-1.4%	9.9%
Contentree Joongang	036420.KS	10/18/2019	34.3%	9.7%	10.9%	26.6%	-25.0%	32.7%	9.3%	6.0%	17.5%	-38.6%
MCNEX	097520.KS	6/6/2021	8.3%	-2.2%	0.6%	-4.7%	-20.4%	15.9%	-0.9%	-1.4%	-3.5%	-2.8%
PI Advanced Material	178920.KS	8/9/2021	-17.1%	-5.4%	6.1%	-21.8%	-32.3%	-15.6%	-4.1%	9.1%	-12.8%	-9.0%
LX Semicon	108320.KS	11/3/2022	19.6%	-0.5%	7.3%	-3.9%	NA	12.6%	4.7%	2.7%	-9.0%	NA
SK Ocean Plant	100090.KS	4/19/2023	-8.9%	-11.8%	-9.7%	-4.6%	NA	-1.4%	-4.2%	-8.2%	-5.9%	NA
BH	090460.KS	6/20/2023	-19.9%	-12.7%	-3.6%	NA	NA	-12.7%	-10.1%	-3.7%	NA	NA

Note: Share performances are based on adjusted share price. Source: WiseFN, Morgan Stanley Research

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